

The free Boson: $\varphi(x^0, x^1)$ be real massless scalars defined on $S^1 \times \mathbb{R}$, $x^0, x^1 \in \mathbb{R}$

$$x^1 \sim x^1 + 2\pi$$

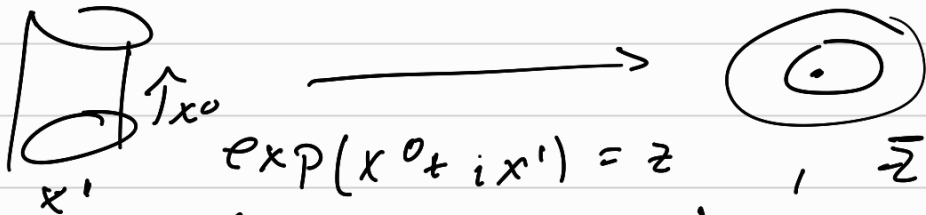
$$\rightarrow S = \frac{1}{4\pi\alpha'} \int dx^0 dx^1 \sqrt{|h|} h^{\alpha\beta} \partial_\alpha \varphi \partial_\beta \varphi$$

Here we have $|h| = \det(h_{\alpha\beta})$ and choose $S^1 \times \mathbb{R}$ to be $\subset \mathbb{R}^2$ to inherit $h_{\alpha\beta} = (\pm 1, 1)$

$$\leadsto S = \frac{1}{4\pi\alpha'} \int dx^0 dx^1 [(\partial_0 \varphi)^2 + (\partial_1 \varphi)^2]$$

K : normalization constant.

• String worldsheet action:



$$\exp(x^0 + ix^1) = z, \quad \bar{z} = e^{x^0 - ix^1}$$

• change of variables:

$$dx^0 = \frac{dx^0}{dz} dz + \frac{dx^0}{d\bar{z}} d\bar{z}$$

$$\begin{cases} \rho_1(z) = x^0 + ix^1, & \rho_1(\bar{z}) = x^0 - ix^1 \\ x^0 = \frac{1}{2}(\rho_1(z) + \rho_1(\bar{z})) \\ \frac{1}{2i}(\rho_1(z) - \rho_1(\bar{z})) = x^1 \end{cases}$$

$$dx^0/dz = \frac{1}{2} \frac{1}{z} \bar{z} = \frac{1}{2z}$$

$$\partial x^0 / \partial \bar{z} = \frac{1}{2} \bar{z}^{-1}$$

$$\partial x^1 / \partial z = \frac{1}{2i} \left(\frac{\bar{z}}{z} \right) \frac{1}{z} = \frac{1}{2iz}$$

$$\partial x^1 / \partial \bar{z} = \frac{1}{2i} \left(\frac{\bar{z}}{z} \right) \frac{z}{\bar{z}} = (-1/2i) \bar{z}$$

$$dx^0 = \frac{1}{2z} dz + \frac{1}{2} \frac{1}{\bar{z}} d\bar{z}$$

$$dx^1 = \frac{1}{2iz} dz - \frac{1}{2i} \bar{z} d\bar{z}$$

$$dx^0 \wedge dx^1 = \left(\frac{1}{2z} dz + \frac{1}{2} \frac{1}{\bar{z}} d\bar{z} \right) \left(\frac{1}{2iz} dz - \frac{1}{2i} \bar{z} d\bar{z} \right)$$

$$= \frac{1}{4iz^2} dz \wedge d\bar{z} - \frac{1}{4i\bar{z}^2} d\bar{z} \wedge dz$$

$$= \frac{1}{2iz^2} dz \wedge d\bar{z}$$

$$\leadsto g_{\alpha\beta} = \begin{pmatrix} 0 & (2z\bar{z})^{-1} \\ (2z\bar{z})^{-1} & 0 \end{pmatrix}, \quad g^{\alpha\beta} = \begin{pmatrix} 0 & 2z\bar{z} \\ 2z\bar{z} & 0 \end{pmatrix}$$

$$\sqrt{|g|} = \frac{1}{4|z|^4} = \frac{1}{2|z|^2}$$

So: area element has a factor of $\frac{1}{2|z|^2}$ here

$$\Rightarrow \frac{1}{2|z|^2} = \frac{1}{\sqrt{g}} \Rightarrow \sqrt{g} dz d\bar{z}$$

$$\bullet S = \frac{1}{4\pi\alpha'} \int dz d\bar{z} \partial\varphi \cdot \bar{\partial}\varphi$$

$$\text{E.O.M: } 0 = \delta_\varphi S$$

$$\begin{aligned} 0 &= \frac{1}{4\pi\alpha'} \int dz d\bar{z} (\partial\delta\varphi \cdot \bar{\partial}\varphi + \partial\varphi \cdot \bar{\partial}\delta\varphi) \\ &= \frac{1}{4\pi\alpha'} \int dz d\bar{z} (\partial(\delta\varphi \bar{\partial}\varphi) - \delta\varphi \cdot \partial\bar{\partial}\varphi + \bar{\partial}(\partial\varphi \cdot \delta\varphi) \\ &\quad - \bar{\partial}\partial\varphi \cdot \delta\varphi) \\ &= -\frac{1}{2\pi\alpha'} \int dz d\bar{z} (\bar{\partial}\partial\varphi) \delta\varphi \end{aligned}$$

$$\text{E.O.M: } \boxed{\partial\bar{\partial}\varphi = 0}$$

$\leadsto$ Two vanishing currents:

$$\partial j(z) = 0$$

$$\bar{\partial} \bar{j}(\bar{z}) = 0$$

$$j(z) = i\partial\varphi(z, \bar{z})$$

(Chiral)

$$\bar{j}(\bar{z}) = i\bar{\partial}\varphi(z, \bar{z})$$

(Anti-chiral)

$\bullet$ Conformal properties: $\omega(z)$ $\varphi(z, \bar{z}) = \varphi(\omega, \bar{\omega})$

$$\begin{aligned} S &\rightarrow \frac{1}{4\pi\alpha'} \int dz d\bar{z} \partial_z \varphi'(z, \bar{z}) \cdot \bar{\partial}_{\bar{z}} \varphi'(z, \bar{z}) \\ &= \frac{1}{4\pi\alpha'} \int d\omega \left(\frac{dz}{d\omega}\right) d\bar{\omega} \left(\frac{d\bar{z}}{d\bar{\omega}}\right) \underbrace{\frac{\partial\omega}{\partial z} \partial_\omega \varphi(\omega, \bar{\omega}) \bar{\partial}_{\bar{\omega}} \varphi(\omega, \bar{\omega}) \left(\frac{\partial\bar{\omega}}{\partial\bar{z}}\right)} \\ &= \frac{1}{4\pi\alpha'} \int d\omega d\bar{\omega} \partial_\omega \varphi(\omega, \bar{\omega}) \bar{\partial}_{\bar{\omega}} \varphi(\omega, \bar{\omega}) \end{aligned}$$

so φ has conformal weight 0

$$\varphi: (h_\varphi, \bar{h}_\varphi) = (0, 0)$$

and from the above we know

$$j(z): (h_j, \bar{h}_j) = (1, 0)$$

$$\bar{j}(\bar{z}): (h_{\bar{j}}, \bar{h}_{\bar{j}}) = (0, 1)$$

Two point f^{us} and Laurent Mode Alg.:
(Green's f^u)

$$\begin{aligned} G(z, \bar{z}, \omega, \bar{\omega}) &\equiv \langle \varphi(z, \bar{z}) \varphi(\omega, \bar{\omega}) \rangle = \langle 0 | R(\varphi(z, \bar{z}) \varphi(\omega, \bar{\omega})) | 0 \rangle \\ &= \int D\varphi e^{-S[\varphi]} \varphi(z, \bar{z}) \varphi(\omega, \bar{\omega}) \end{aligned}$$

$$\varphi(z, \bar{z}) = \int \frac{d\omega d\bar{\omega}}{(2\pi)^2} G(z, \bar{z}, \omega, \bar{\omega}) f(\omega, \bar{\omega})$$

$$\partial\bar{\partial}\varphi = \int d\omega d\bar{\omega} \partial\bar{\partial} G f$$

$$\partial\bar{\partial}P = f(z, \bar{z}) = \int \frac{\partial^2 \psi \partial \bar{\psi}}{\partial z \partial \bar{z}} \delta(z-u) \delta(\bar{z}-\bar{u}) f(u, \bar{u})$$

$$\therefore \oint \frac{d\omega}{2\pi i} \delta(z-\omega) f(\omega) = f(z)$$

$$\text{But } f^{(n)}(z) = \frac{1}{2\pi i} \oint \frac{f(\omega)}{(z-\omega)^{n+1}}$$

$$f^{(0)}(z) = \frac{1}{2\pi i} \oint \frac{f(\omega)}{(z-\omega)^1} d\omega$$

Thus, complex delta f^h

$$\delta(z-u) = \left(\frac{1}{2\pi i}\right) \frac{1}{z-u}$$

$\sim (-2\pi)$ comes from the normalization $\frac{1}{2\pi} \int d\psi d\bar{\psi} dz d\bar{z}$

$\therefore$

$$\partial\bar{\partial}G(z, \bar{z}, u, \bar{u}) = -2\pi \delta^{(2)}(z-u)$$

$$\text{Proof } \int D\psi \frac{\delta}{\delta\psi(z, \bar{z})} \{ e^{S[\psi]} \psi(u_1, \bar{u}_1) \dots \psi(u_n, \bar{u}_n) \} = 0$$

$$\int D\psi \left(- \int d\omega^2 \left(\partial(\delta(z-u)) \bar{\partial}\psi + \partial\psi \bar{\partial}\delta(z-u) \right) \psi \dots \right. \\ \left. + \int D\psi e^{S[\psi]} \sum_{i=1}^n (\delta(z-u_i) \psi(u_1, \dots) \dots \psi(u_i, \bar{u}_i) \dots) \right)$$

$$\frac{1}{2\pi} (\partial\bar{\partial}\psi)(z) \psi(u_1, \bar{u}_1) \psi(u_2, \bar{u}_2) = \delta(z-u_1) \psi_2 +$$

$\dots$ not too sure $\dots$

$\leadsto$ Going back to

$$\partial\bar{\partial}G(z, \bar{z}, u, \bar{u}) = -2\pi \delta^{(2)}(z-u)$$

$$2\pi \delta^{(2)} = \partial_z \bar{z}^{-1} \mid \delta(z) = \frac{1}{2\pi i z} =$$

$$\partial_z \frac{1}{(\bar{z}-\bar{u})} = 2\pi \delta^{(2)}(z-u)$$

$$\Rightarrow \partial(\bar{\partial}G(z, \bar{z}, u, \bar{u})) + \frac{1}{\bar{z}-\bar{u}} = 0$$

$$\bar{\partial}G(z, \bar{z}, u, \bar{u}) = -\frac{1}{(\bar{z}-\bar{u})}$$

$$\int d\bar{z} \bar{\partial}G = -\oint (\bar{z}-\bar{u})^{-1}$$

$$G(z, \bar{z}, u, \bar{u}) = -\ln|\bar{z}-\bar{u}| + f(z)$$

$$\bar{\partial}f = 0$$

$$\bar{\partial}(G + \frac{1}{\bar{z}-\bar{u}}) \Rightarrow G(\bar{z}) = -\ln|\bar{z}-\bar{u}| + g(\bar{z})$$

and so we find

$$G(z, \bar{z}, u, \bar{u}) = -\ln|\bar{z}-\bar{u}| - \ln|z-u|$$

$$G(z, \bar{z}, w, \bar{w}) = -\ln|z-w|^2$$

* which we see: $\bar{\partial}_z \frac{1}{z-\bar{z}} = 2\pi \delta^2(z-\bar{z})$

$$\frac{1}{2\pi i} \int_D dz d\bar{z} f(z) (\bar{\partial}_z \frac{1}{z-\bar{z}}) = \oint_{\partial D} f(z) (\frac{1}{z-\bar{z}}) = f(\bar{z})$$

* Now, note: Wick's thm:

$$T(\phi(x_1) \dots \phi(x_n)) = : \phi_1 \dots \phi_n : + (\text{all possible contractions})$$

ex:

$$T(\phi_1 \phi_2) = : \phi_1 \phi_2 : + \langle 0 | T \{ \phi_1 \phi_2 \} | 0 \rangle$$

→ Can also have fields which already hav. a normal ordering

$$T[\phi_1 \dots : \phi_p \dots \phi_q \dots \phi_r \dots]$$

Modification: Do not include two pt Green fns of fields which were originally in the same normal ordered piece.

$$A(z) = A[\varphi](z), \quad B(w) = B[\varphi](w)$$

⇒

$$\varphi(z, \bar{z}) \varphi(w, \bar{w}) = -\ln|z-w|^2 + : \varphi(z, \bar{z}) \varphi(w, \bar{w}) :$$

$$\text{OPE: } \varphi(z, \bar{z}) \varphi(w, \bar{w}) \sim -\ln|z-w|^2 + \dots$$

⌈ non-singular terms as

$z \rightarrow w$

• Free Boson is not "quasi-primary field"

→ E.M. tensor: can find it via

$$\delta \varphi = \kappa^\alpha \partial_\alpha \varphi \quad \text{in } S \quad \text{and solve for } \partial_\alpha \kappa^\alpha$$

or:

$$T_{ab} = \frac{1}{4\pi} \kappa^\gamma \frac{1}{|\gamma|} \frac{\delta S}{\delta g^{ab}}$$

$$\Rightarrow T_{zz} = \gamma \partial \varphi \partial \varphi, \quad T_{\bar{z}\bar{z}} = \gamma \bar{\partial} \varphi \bar{\partial} \varphi$$

Quantum:

$$T(z) = : \gamma \partial \varphi \partial \varphi : (z)$$

* Fix γ by requiring $j(z)$ be a primary field of conf.

dim $h=1$ wrt. $T(z)$.

$$h = I \text{ in West and } \gamma = 1/2$$

West keeps (- sign)

$$\begin{cases} T(z) = -\frac{1}{2} : \partial \phi \partial \phi : (z) \\ \bar{T}(\bar{z}) = -\frac{1}{2} : \bar{\partial} \phi \bar{\partial} \phi : (\bar{z}) \end{cases}$$

$\therefore$

$$T(z) \phi(w, \bar{w}) = -\frac{1}{2} : \partial \phi \partial \phi : (z) \phi(w, \bar{w})$$

$$T(\quad) = -\frac{1}{2} : \partial \phi(z) \partial \phi(z) \phi(w, \bar{w}) : \\ + - \langle 0 | T \{ \partial \phi \phi \} | 0 \rangle : \partial \phi :$$

• Note the normal ordering of $: \partial \phi : = \partial \phi$

Thus

$$T(z) \phi(w, \bar{w}) = - \langle \partial \phi(z) \phi(w) \rangle - \frac{1}{2} : \partial \phi(z) \partial \phi(z) \phi(w, \bar{w}) :$$

However, we already know, by OPE that

$$T(z) \phi(w) = \frac{h \phi(w)}{(z-w)^2} + \frac{\partial \phi(w)}{(z-w)}$$

$$h_\phi = 0 \text{ so } \partial \phi(w)$$

$$T(z) \phi(w, \bar{w}) = \frac{\partial \phi(w)}{(z-w)}$$

Moreover, let us work this out to prove $h_\phi = 0$:

$$- \langle \phi(z) \phi(w) \rangle = \ln |z-w|^2$$

$$- \partial_z \langle \phi(z) \phi(w) \rangle = - \langle \partial_z \phi(z) \phi(w) \rangle = \partial_z \ln(z-w) \\ = \frac{1}{z-w}$$

$$\therefore T(z) \phi(w, \bar{w}) = \frac{\partial \phi(w)}{z-w}$$

$$\leadsto \partial \bar{\partial} \phi(z, \bar{z}) = 0$$

$$\partial(\bar{\partial} \phi) \Rightarrow \bar{\partial} \psi(z, \bar{z}) = g(\bar{z})$$

$$\bar{\partial}(\partial \phi) \Rightarrow \partial \psi(z, \bar{z}) = f(z)$$

$$\therefore \psi(z, \bar{z}) = f(z) + g(\bar{z})$$

$$\text{or } \boxed{\psi(z, \bar{z}) = \psi_L(z) + \psi_R(\bar{z})}$$

$$\bullet \psi(z, \bar{z}) = \sum_n z^{-n} (\psi_L)_n + \sum_m \bar{z}^{-m} (\psi_R)_m$$

$$\partial\phi = \sum_n (-n) z^{-n-1} (\phi_L)_n$$

But recall that

$$j(z) = i\partial\phi = \sum_n j_n z^{-n-1}$$

$$\bar{j}(\bar{z}) = i\bar{\partial}\phi = \sum_n \bar{j}_n \bar{z}^{-n-1}$$

$$\therefore \int_0^1 i\partial\phi dz = \int_0^1 dz \sum_n j_n z^{-n-1}$$

$$i\phi(z, \bar{z}) = j_0 \ln(z) - \sum_{n \neq 0} j_n \left(\frac{1}{n}\right) z^{-n} + \phi_0$$

+ Anti chiral

$$\phi(z, \bar{z}) = \tilde{\phi}_0 - i(j_0 \ln(z) + \bar{j}_0 \ln(\bar{z})) + i \sum_{n \neq 0} \frac{1}{n} (j_n z^{-n} + \bar{j}_n \bar{z}^{-n})$$

~> If we let $\bar{z} = z^*$ (C.C.), the ϕ is defined on $\mathbb{C}$. As such it must be invariant under $z \mapsto e^{2\pi i} z$ (Rotations) Since all states are eq. on $|z\rangle$.

$$z \rightarrow e^{2\pi i} z$$

$$\bar{z} \rightarrow \bar{e}^{2\pi i} \bar{z}$$

$$\phi(z, \bar{z}) = \tilde{\phi}_0 - i(j_0 \ln(e^{2\pi i} z) + \bar{j}_0 \ln(\bar{e}^{2\pi i} \bar{z})) + \dots$$

$$\sim \frac{1}{e^{2\pi i n}} = 1 \text{ so last term is invariant:}$$

$$-i(j_0 2\pi i - \bar{j}_0 2\pi i) - i(j_0 \ln(z) + \bar{j}_0 \ln(\bar{z})) + \dots$$

$$\Rightarrow j_0 = \bar{j}_0$$

$$\phi_0 = \phi_L + \phi_R$$

$$\therefore \phi(z, \bar{z}) = \phi_L - i j_0 \ln(z) + i \sum_{n \neq 0} \frac{1}{n} j_n z^{-n} + \phi_R - i \bar{j}_0 \ln(\bar{z}) + i \sum_{n \neq 0} \frac{1}{n} \bar{j}_n \bar{z}^{-n}$$

$$= \phi_L(z) + \phi_R(\bar{z})$$

$$j_0 = \bar{j}_0$$

$$\text{where West calls } j_0 = p_L, \bar{j}_0 = p_R$$

$$\sim \phi \equiv \phi_0, \phi_L = \phi/2 = \phi_R$$

$$p \equiv p_R = p_L \text{ or } p = j_0$$

* West uses

$$\alpha_n = j_n$$



Take $n \rightarrow -n : -i \sum_{n \neq 0} \alpha_{-n} z^n$

$$\partial\varphi = \partial\varphi_L = -i \sum_{n=-\infty}^{\infty} \alpha_{-n} z^{n-1}$$

$$\bar{\partial}\varphi = \bar{\partial}\varphi_R = -i \sum \bar{\alpha}_{-n} \bar{z}^{n-1}$$

we conclude:

$$\left\{ \begin{array}{l} \langle \varphi_L \varphi_L \rangle = -\ln(z-w) \\ \langle \varphi_R \varphi_R \rangle = -\ln(\bar{z}-\bar{w}) \end{array} \right. \quad \langle \varphi_L \varphi_R \rangle = 0$$

$$\text{so } \varphi(z) \varphi(w) = -\ln|z-w| - \ln|\bar{z}-\bar{w}| + :\varphi(z) \varphi(w):$$

$$\partial_z \partial_w () = \frac{1-1}{(z-w)^2} + :\partial_z \varphi(z, \bar{z}) \partial_w \varphi(w, \bar{w}):$$

$$\partial_z \varphi(z) = \left(\sum_{n=0}^{\infty} \frac{(z-w)^n}{n!} \partial^n (\partial\varphi)(w) \right)$$

$$\partial_z \varphi(z) \partial_w \varphi(w) = -\frac{1}{(z-w)^2} + \sum_{n=0}^{\infty} \frac{(z-w)^n}{n!} :\partial^{n+1} \varphi(w) \partial\varphi(w):$$

$\Rightarrow$ OPE

$$\partial\varphi \partial_w \varphi = \frac{-1}{(z-w)^2} + \dots$$

similar for $\bar{\partial} \bar{\partial}$.

$$\begin{aligned} * \partial\varphi = \partial\varphi_L = -i \sum \alpha_{-n} z^{n-1} \quad \text{or} \\ \oint \frac{dz}{2\pi i} z^{-m} i \partial\varphi = \oint \frac{dz}{2\pi i} \alpha_{-n} z^{n-1-m} \\ n-1-m = -1 \Rightarrow n=m \end{aligned}$$

$$\therefore \alpha_n = \oint_0 \frac{dz}{2\pi i} z^n i \partial\varphi$$

$$\begin{aligned} \therefore [\alpha_n, \alpha_m] &= - \oint \frac{d\zeta}{2\pi i} \oint \frac{dz}{2\pi i} \zeta^m z^n \overbrace{\partial\varphi(z) \partial\varphi(\zeta)} \\ &= \oint \frac{d\zeta}{2\pi i} \zeta^m \oint \frac{dz}{2\pi i} z^n \frac{1}{(z-\zeta)^2} = \oint \frac{d\zeta}{2\pi i} \zeta^m n \zeta^{n-1} \\ &\quad m = -n \end{aligned}$$

$$= \delta_{m+n} n$$

$$\Rightarrow [\alpha_n, \alpha_m] = n \delta_{n+m, 0} \quad [\alpha_n, \bar{\alpha}_m] = n \delta_{n+m, 0}$$

$$[\alpha_n, \bar{\alpha}_m] = 0$$

$$[q, p] = i$$

$$p = \alpha_0$$

* Centre of mass momentum $p = \frac{1}{4\pi} \int_0^{2\pi} dx' \frac{\phi(x^0, x')}{\partial(-ix^0)} = \frac{j_0 + \bar{j}_0}{2}$
 $= j_0$

$$\phi_0 = \frac{1}{2\pi} \int_0^{2\pi} dx' \phi(x^0=0, x') = q \quad (\text{C.O.M. of string})$$

where is where $[q, p] = i$ comes from

$$\Rightarrow \frac{1}{4\pi} \int d\bar{\sigma} d\sigma \partial\phi \bar{\partial}\phi = \frac{1}{8\pi i} \int d\tau d\sigma (-\partial_\tau \phi \partial_\sigma \phi + \partial_\sigma \phi \partial_\tau \phi)$$

used a Wick rotation $x^0 = i\tau$

* Conjugate momentum

$$\Pi_\phi = \frac{\partial \mathcal{L}}{\partial \dot{\phi}} = -\frac{1}{4\pi i} \partial_\tau \phi$$

Equal time Com. - relations (equal time)

$$[p(\tau, \sigma), \pi(\tau, \sigma')] = i \delta(\sigma - \sigma')$$

$$[\phi(\tau, \sigma), \partial_\tau \phi(\tau, \sigma')] = 4\pi \delta(\sigma - \sigma')$$

From E.O.M: and invariance $\bar{\sigma} \rightarrow \bar{\sigma} + 2\pi i$ from $\sigma \rightarrow \sigma + 2\pi$

$$\phi(\tau, \sigma) = q(\tau) + i \sum_{n \neq 0} \frac{\alpha_n(\tau)}{n} e^{-in\sigma}$$

(can just expand the expression above)

$$z = e^{x^0 + ix^1}$$

$$\Rightarrow \phi_L = q_L - i j_0 \ln(e^{x^0 + ix^1}) - i \sum_{n \neq 0} \frac{\alpha_{-n}}{n} e^{n(x^0 + ix^1)}$$

$$x^0 = i\tau \quad (\text{Wick})$$

$$x^1 = \sigma$$

$$\Rightarrow \phi_L = q_L - i j_0 (i)(\tau + \sigma) - i \sum_{n \neq 0} \frac{\alpha_n}{n} e^{-in(\tau - \sigma)}$$

Back to PIP²:

$$\lim_{z \rightarrow 0} (\partial\phi | 0 \rangle) = -i \sum_{n \in \mathbb{Z}} \alpha_n z^{n-1} | 0 \rangle$$

$$\alpha_n | 0 \rangle = 0, \quad n \geq 0$$

$$:\alpha_n \alpha_m: = \begin{cases} \alpha_n \alpha_m & \text{if } m > n \\ \alpha_m \alpha_n & \text{if } n > m \end{cases}$$

$$\Rightarrow T(z) T(w) = \frac{1}{z^2} \cdot \partial\phi^2(z) \cdot \partial\phi^2(w) \cdot$$

$$\begin{aligned}
 &= : \partial \phi(z) \partial \phi(w) : \langle \partial \phi(z) \partial \phi(w) \rangle + \frac{1}{2} \langle \partial \phi(z) \partial \phi(w) \rangle^2 \\
 &\quad + : (\partial \phi)^2(z) (\partial \phi)^2(w) : \\
 &= : \partial \phi(z) \partial \phi(w) : \left(-\frac{1}{(z-w)^2} \right) + \frac{1}{2} \left(-\frac{1}{(z-w)^2} \right)^2
 \end{aligned}$$

$$\begin{aligned}
 \text{Recall: } : \partial \phi(z) \partial \phi(w) : &= \sum_{n=0}^{\infty} \frac{(z-w)^n}{n!} \partial^{n+1} \phi(w) \partial \phi(w) \\
 &= : \partial \phi(w) \partial \phi(w) : + : (z-w) \partial^2 \phi(w) \partial \phi(w) : + : \sum_{n=2}^{\infty} \frac{(z-w)^{n-2}}{n!} \partial^{n+1} \phi(w) \partial \phi(w) :
 \end{aligned}$$

Plugging this back in we find, using

$$T(w) = -\frac{1}{2} : \partial \phi \partial \phi :$$

$$\partial T(w) = - : \partial^2 \phi \partial \phi :$$

$$T(z)T(w) = \frac{2T(w)}{(z-w)^2} + \frac{\partial T(w)}{(z-w)} + \frac{1/2}{(z-w)^4} + \dots$$

* Conclude that T has conf. weight $h=2$,
and the free Boson has central charge
 $C=1$

$$T(z) = -\frac{1}{2} : \partial \phi(w) \partial \phi(w) : = \sum_{n=-\infty}^{\infty} z^{-n-2} L_n$$

$$(LHS) = -\frac{1}{2} : (-i)^2 \sum_{n=-\infty}^{\infty} \alpha_n z^{-n-1} \sum_{m=-\infty}^{\infty} \alpha_m z^{-m-1} :$$

$$= \frac{1}{2} \sum_n \sum_m z^{-n-m-2} : \alpha_n \alpha_m :$$

$$= \sum_{k=-\infty}^{\infty} z^{-k-2} \left(\frac{1}{2} \sum_n : \alpha_n \alpha_{k-n} : \right)$$

$$\therefore L_n = \frac{1}{2} : \sum_m \alpha_{n-m} \alpha_m :$$

Given a two operators $A(z)B(w)$ we can define the normal ordering by the meromorphic portion of the OPE

$$: \partial \phi(z) \partial \phi(w) : = \lim_{z \rightarrow w} \left(\partial \phi(z) \partial \phi(w) - \left(-\frac{1}{(z-w)^2} \right) \right)$$

Vertex Operators:

→ As we just saw: $\phi(z, \bar{z})$ is not a conformal field
 Since $h = \bar{h} = 0$ and its OPE is not a meromorphic $\oint \gamma$.

• We can build non-zero conf. fields with $\phi(z, \bar{z})$

Def: The Vertex Operator $V(z, \bar{z})$ is a primary field of
 conf. dimension $(h, \bar{h}) = (\frac{k^2}{2}, \frac{k^2}{2})$ wrt. the E.M.
 tensor $T(z) = \frac{1}{2} \dot{\phi} \dot{\phi}$ and $\bar{T}(\bar{z}) = \frac{1}{2} \dot{\bar{\phi}} \dot{\bar{\phi}}$:
 $V(z, \bar{z}) = \exp(i k \phi(z, \bar{z}))$

→ Note West uses $\phi_L(z)$ to sep. V into its chiral and
 anti-chiral parts:

$$\Rightarrow e^{i k \phi_L(z)} = e^{i k \phi_L(z)} e^{i k q} e^{k \cdot p \phi_L(z)} e^{i k \phi_R(z)}$$

where we break down $\phi_L(z)$

$$\phi_L(z) = q - i p \phi_L(z) - i \sum_{n=0}^{\infty} \frac{\alpha_{-n}}{n} z^n$$

$$= -i \sum_{n=0}^{\infty} \frac{\alpha_{-n}}{n} z^n + q - i p \phi_L(z) - i \sum_{n=0}^{\infty} \frac{\alpha_{-n}}{n} z^n$$

$$\hookrightarrow (-n) \rightarrow n$$

$$= \underbrace{i \sum_{n=1}^{\infty} \frac{\alpha_n}{n} z^{-n}}_{\phi_{L>}(z)} + \underbrace{(q - i p \phi_L(z))}_{\phi_{L0}(z)} - \underbrace{i \sum_{n=1}^{\infty} \frac{\alpha_{-n}}{n} z^n}_{\phi_{L<}(z)} = \phi_L(z)$$

→ Take $[q, p] = i$

$$\Rightarrow \partial \phi_L(z) : e^{i k \phi_L(w)} : = \partial \phi_L(z) : \sum_{n=0}^{\infty} \frac{(i k \phi_L(w))^n}{n!} :$$

$$\partial \phi_L(z) : \phi_L(w)^n : = : \partial \phi_L(z) \phi_L(w)^n : + n \langle \partial \phi_L(z) \phi_L(w) \rangle : \phi_L(w)^{n-1} :$$

$$\langle \phi_L(z) \phi_L(w) \rangle = -\ln|z-w|$$

$$\Rightarrow \langle \partial \phi_L(z) \phi_L(w) \rangle = -\frac{1}{z-w}$$

$$\Rightarrow \partial \phi_L(z) : \phi_L(w)^n : = : (\text{Non singular}) : - \frac{n}{z-w} : \phi_L(w)^{n-1} :$$

$$\text{Plugging this back in } \Rightarrow \partial \phi_L(z) : e^{i k \phi_L(w)} : = -\frac{1}{z-w} \sum_{n=0}^{\infty} \frac{(i k)^n}{(n-1)!} : \phi_L(w)^{n-1} :$$

where $n=0$ is no longer defined, hence

$n=0$ is zero

$$\partial \varphi_L(z) : e^{ik\varphi_L(w)} : = -\frac{1}{z-w} \sum_{n=1}^{\infty} \frac{(ik)(ik)^{n-1}}{(n-1)!} : \varphi_L(w)^{n-1} : + \dots$$

and finally

$$\partial \varphi_L(z) : e^{ik\varphi_L(w)} : = \frac{-ik}{(z-w)} e^{ik\varphi_L(w)} + \dots$$

where $e^{ik\varphi_L(w)}$ is normal ordered by our prescription
 $\varphi_L(z) = \varphi_{L-}(z) + \varphi_{L0}(z) + \varphi_{L+}(z)$

* Next: $T(z) : e^{ik\varphi_L(w)} :$, with $T(z) = -\frac{1}{2} : \partial \varphi(z) \partial \varphi(z) :$

$$T(z) : e^{ik\varphi_L(w)} : = -\frac{1}{2} : \partial \varphi(z) \partial \varphi(z) : e^{ik\varphi_L(w)} :$$

$$+ \sum_{n=0}^{\infty} \frac{(ik)^n}{n!} -\frac{1}{2} : \partial \varphi(z) \partial \varphi(z) : : \varphi_L^n(w) :$$

$$- \frac{1}{2} \left(\frac{-ik}{z-w} \right) : \partial \varphi(z) : e^{ik\varphi_L} : - n(n-1) \frac{1}{2} \left(\frac{-ik}{z-w} \right)^2 : \varphi_L(w)^{n-2} :$$

→ minus sign from $T(z) = -\frac{1}{2} : \partial \varphi \partial \varphi :$

(Second term):

$$\left(\frac{-ik}{z-w} \right) \sum_{n=2}^{\infty} \frac{(ik)^{n-2}}{(n-2)!} : \varphi_L(w)^{n-2} : = \frac{(k^2/2)}{(z-w)^2} : e^{ik\varphi_L(w)} :$$

(first term) $\frac{1}{(z-w)} : (ik \partial_z \varphi_L(z) e^{ik\varphi_L(w)} :)$

expand near $w \rightarrow z$

Taylor expand $\partial_z \varphi_L(z)$ around w .

$$\partial_z \varphi_L(z) = \sum_{n=0}^{\infty} \frac{\partial^n (\partial \varphi)(w)}{n!} (z-w)^n$$

here $n=0$ will yield term not $\propto z$:

$$\left(\frac{1}{z-w} \right) : ik \partial \varphi(w) e^{ik\varphi_L(w)} : + : ik \sum_{n=1}^{\infty} \frac{\partial^n (\partial \varphi)(w)}{n!} (z-w)^n e^{ik\varphi_L(w)} :$$

$$\therefore \left(\frac{1}{z-w} \right) : \partial e^{ik\varphi_L(w)} : + \dots$$

$$T(z) : e^{ik\varphi_L(w)} : = \frac{(k^2/2)}{(z-w)^2} : e^{ik\varphi_L(w)} : + \frac{1}{z-w} \partial e^{ik\varphi_L(w)} + \dots$$

Vertex op. has conformal weights

$$(h, \bar{h}) = (k^2/2, k^2/2)$$

as claimed. $\square$

Claim:
$$:e^{ik_1\phi_L(z)}: :e^{ik_2\phi_L(w)}: = (z-w)^{k_1 k_2} :e^{ik_1\phi_L(z) + ik_2\phi_L(w)}:$$

Proof:

$$e^A e^B = e^B e^A e^{[A,B]}$$

$$\leadsto :e^{ik_1\phi_L(z)}: = e^{ik_1\phi_L(z)} e^{ik_1 q_1} e^{ik_1 p_1(z)} e^{ik_1\phi_L(z)}$$

So all we need to commute is

$$e^{ik_1\phi_L(z)} e^{ik_2\phi_L(w)}$$

$$\Rightarrow A = ik_1\phi_L(z) = -k_1 \sum_{n=1}^{\infty} \frac{\alpha_n}{n} z^{-n}$$

$$B = ik_2\phi_L(w) = +k_2 \sum_{n=1}^{\infty} (\alpha_{-n}/n) w^n$$

$$\Rightarrow [A, B] = -k_1 k_2 \sum_{n,m} \frac{1}{nm} w^m z^{-n} [\alpha_n, \alpha_{-m}]$$

$$= -k_1 k_2 \sum_{n,m} \frac{1}{nm} \delta_{n-m,0} w^m z^{-m}$$

$$= -k_1 k_2 \sum_{n=1}^{\infty} \frac{1}{n} \left(\frac{w}{z}\right)^n$$

Since $\ln(1-x) = -\sum_{n=1}^{\infty} \frac{x^n}{n}$

$$[A, B] = k_1 k_2 \ln\left(1 - \frac{w}{z}\right) \quad (\text{as claimed in the text})$$

$\leadsto$ need to move p_1 past q_2

$$A' = k_1 p_1(z)$$

$$B' = ik_2 q_2$$

$$[A, B]' = ik_2 k_1 \ln(z) [p, q] = k_1 k_2 \ln(z)$$

$$[A, B]_{\text{total}} = k_1 k_2 \ln(z-w)$$

$$:e^{ik_1\phi_L(z)}: :e^{ik_2\phi_L(w)}: = (z-w)^{k_1 k_2} :e^{ik_1\phi_L(z) + ik_2\phi_L(w)}: \quad \square$$

* We can generalize this result:

$$\prod_{i=1}^n :e^{ik_i\phi_L(z_i)}: = \prod_{\substack{i,j=1 \\ i < j}}^n (z_i - z_j)^{k_i k_j} : \exp \sum_{\ell=1}^n ik_{\ell}\phi_L(z_{\ell}) :$$

• Consider a DIFFERENT scalar field ϕ :

(i) ϕ has the same OPE

$$\phi(z, \bar{z}) \phi(w, \bar{w}) = -\ln(z-w) - \ln(\bar{z}-\bar{w}) + \dots$$

• However its E.M. tensor is given by

$$\begin{cases} T = -\frac{1}{2} \partial \phi \partial \phi = -Q \partial^2 \phi \\ \bar{T} = -\frac{1}{2} \bar{\partial} \phi \bar{\partial} \phi = -Q \bar{\partial}^2 \phi \end{cases}$$

Q: Background charge $\in \mathbb{C}$

• We can find the central charge c_ϕ from TT and the conformal factors of ϕ from $T\phi$:

(i) Central charge c_ϕ :

$$\begin{aligned} T(z)T(w) &= (\partial \phi(z) \partial \phi(z) + Q \partial^2 \phi(z)) (\partial_\omega \phi \partial_\omega \phi + Q \partial_\omega^2 \phi) \\ &= \frac{1}{4} \partial_z^2 \phi \partial_z \phi : \partial_\omega \phi \partial_\omega \phi + \frac{1}{2} \partial_z^2 \phi \partial_\omega \phi \partial_\omega \phi Q + \frac{1}{2} \partial_z \phi \partial_z \phi Q \partial_\omega^2 \phi \\ &\quad + Q^2 \partial_z^2 \phi \partial_\omega^2 \phi \end{aligned}$$

$$(1^{st}): \frac{1}{2} \langle \partial_z \phi \partial_\omega \phi \rangle : \partial_z \phi \partial_\omega \phi : + \frac{1}{4} : \partial_z \phi \partial_z \phi \partial_\omega \phi \partial_\omega \phi : + \frac{1}{4} \langle \partial_z \phi \partial_\omega \phi \rangle^2$$

$$= \frac{1}{2} \frac{1}{(z-w)^4} - \frac{: \partial_\omega \phi \partial_\omega \phi :}{(z-w)^2} - \frac{1}{2} \partial : \frac{\partial_\omega \phi \partial_\omega \phi}{(z-w)} + \dots$$

$$(2^{nd}): \frac{1}{2} \partial_z^2 \phi : \partial_\omega \phi \partial_\omega \phi :$$

$$= \frac{1}{2} (: \partial_z^2 \phi \partial_\omega \phi \partial_\omega \phi :) + \langle \partial_z^2 \phi \partial_\omega \phi \rangle \partial_\omega \phi$$

$$= \partial_z \langle \partial_z \phi \partial_\omega \phi \rangle = -2/(z-w)^3$$

$$- \frac{2 \partial_\omega \phi Q}{(z-w)^3}$$

$$= \frac{2 \partial_\omega \phi Q}{(z-w)^3}$$

$$(3^{rd}) = (1 + 2 \partial_z \phi) / (z-w)^3$$

$$(4^{th}) = \frac{Q^2}{(z-w)^4} \partial_z \partial_\omega \left(\frac{-1}{(z-w)^2} \right) = Q^2 \partial_\omega \left(\frac{2}{(z-w)^3} \right)$$

$$\therefore T(z)T(w) = \frac{1}{2} \frac{(1+12Q^2)}{(z-w)^4} + \frac{2(\partial_\omega \phi - \partial_z \phi)Q}{(z-w)^3} + \dots$$

$\therefore$

Central charge of ϕ : $c_\phi = 1 + 12Q^2$

→ Let us find the conformal factors of ϕ :

$$\begin{aligned}
T(z)\phi(w) &= \left\{ -\frac{1}{2} : \partial_z \phi \partial_z \phi : - Q \partial_z^2 \phi \right\} \phi(w) \\
&= (Reg) - \langle \phi(w) \partial_z \phi(z) \rangle \partial_z \phi - \frac{1}{2} Q \langle \partial_z^2 \phi(z) \phi(w) \rangle \\
&= \frac{\partial_z \phi}{(z-w)} + \frac{1}{2} \frac{Q}{(z-w)^2}
\end{aligned}$$

$$\begin{aligned}
\text{expand } \partial_z \phi &= \sum_{n=0}^{\infty} \frac{(z-w)^n}{n!} \partial^n (\partial \phi)(w) \\
&= \partial_w \phi(w) + (z-w) \partial^2 \phi(w)
\end{aligned}$$

$\leadsto$ Somehow run off by a factor of 2
in $(z-w)^\alpha \rightarrow (z-w)^{\alpha+1}$
which would give (4.1.45)

(TRY) $T(z) \partial \phi(w)$:

$$\begin{aligned}
& -\frac{1}{2} : \partial_z \phi \partial_z \phi : \partial \phi(w) - Q \partial_z^2 \phi \partial_w \phi \\
&= -\langle \partial_z \phi \partial_w \phi \rangle \partial_z \phi - Q \langle \partial_z^2 \phi \partial_w \phi \rangle \\
&= + \frac{1}{(z-w)^2} \partial_z \phi - Q \frac{2}{(z-w)^3} \\
&\text{expand } \partial_z \phi = \sum_{n=0}^{\infty} \frac{(z-w)^n}{n!} \partial^n (\partial \phi)(w) \\
&= \frac{\partial_w \phi}{(z-w)^2} + \frac{\partial^2 \phi}{(z-w)} + \frac{-2Q}{(z-w)^3}
\end{aligned}$$

$\rightarrow D : e^{i\beta\phi(z)}$ is a primary field of conf. weight
 $h_\phi = \frac{1}{2}\beta(\beta - 2iQ)$

* Define the Oscillators by

$$\partial \phi = -i \sum_n \alpha_n z^{-n-1}$$

and so

$$T(z) = -\frac{1}{2} : \partial \phi \partial \phi : - Q \partial^2 \phi = \sum_n L_n z^{-n-2}$$

or

$$L_n = \frac{1}{2} \sum : \alpha_{n-m} \alpha_m : - i(n+1)Q \alpha_n$$

Can obtain this E.M. tensor by the action

$$S = \frac{1}{4\pi\alpha'} \int d^2z \sqrt{|g|} (\partial \phi \bar{\partial} \phi - 2Q R \phi)$$

R : Riemann scalar curvature

$\leadsto$ Expand $g_{\alpha\beta} = \eta_{\alpha\beta} + \mathcal{O}_{\alpha\beta}$
 $\sim \eta_{\alpha\beta} T^{\alpha\beta} + \mathcal{O}(\eta^2)$

Free Fermion: Dirac action w $\eta_{\alpha\beta} = (-1, 1)$

$$S = \frac{1}{4\pi\hbar} \int dx^0 dx^1 \sqrt{|h|} i \bar{\Psi} \gamma^\alpha \partial_\alpha \Psi$$

$$\bar{\Psi} = \Psi^\dagger \gamma^0$$

$$\{\gamma^\alpha, \gamma^\beta\} = -2\eta^{\alpha\beta} \mathbb{1}$$

$$(\gamma^2)_{\alpha\beta} = \sigma^3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

$$(\gamma^0)_{\alpha\beta} = i\sigma^2 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

Wick rotate: $x^0 \rightarrow ix^0$

$$\partial_0 \rightarrow -i\partial_0$$

$$\Rightarrow \Psi^\dagger \gamma^0 \gamma^\mu \partial_\mu \Psi$$

$$\gamma^0 (\gamma^0 (i\partial_0) + \gamma^1 \partial_1)$$

$$\gamma^0 \left(\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \partial_0 + \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \partial_1 \right) = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} 0 & -i\partial_0 + \partial_1 \\ i\partial_0 + \partial_1 & 0 \end{pmatrix}$$

$$= \begin{pmatrix} i\partial_0 + \partial_1 & 0 \\ 0 & -i\partial_0 + \partial_1 \end{pmatrix}$$

$$= 2 \begin{pmatrix} \bar{\partial} & 0 \\ 0 & \partial \end{pmatrix}$$

no Majorana spinor $\Psi_m = \begin{pmatrix} \psi(z, \bar{z}) \\ \bar{\psi}(z, \bar{z}) \end{pmatrix}$

ψ : real field

$$\psi^\dagger = \psi$$

$$\bar{\psi}^\dagger = \bar{\psi}$$

$$\Rightarrow S = \frac{1}{4\pi\hbar} \int dz d\bar{z} \sqrt{|g|} 2 (\psi, \bar{\psi}) \begin{pmatrix} \bar{\partial} & 0 \\ 0 & \partial \end{pmatrix} \begin{pmatrix} \psi \\ \bar{\psi} \end{pmatrix}$$

$$g_{\alpha\beta} = \begin{pmatrix} 0 & 1/2 \\ 1/2 & 0 \end{pmatrix} \quad g^{\alpha\beta} = \begin{pmatrix} 0 & 2 \\ 2 & 0 \end{pmatrix}$$

$$S = -\frac{1}{4\pi} \int dz d\bar{z} \{ \psi \bar{\partial} \psi + \bar{\psi} \partial \psi \}$$

E.O.M.: $\bar{\psi}$: $-\frac{1}{4\pi} \int dz d\bar{z} (\delta\psi \bar{\partial} \psi + \psi \bar{\partial} (\delta\psi))$

Same for ψ

$$\int d^2z (\bar{\partial}(\psi\delta\psi) - \bar{\partial}\psi\delta\psi) + \delta\psi\bar{\partial}\psi$$

Since $-(\bar{\partial}\psi)\delta\psi = \delta\psi\bar{\partial}\psi$

$$\therefore 2\delta\psi\bar{\partial}\psi = 0$$

$$\bar{\partial}\psi = \bar{\partial}\bar{\psi} = 0$$

Thus, the two fermionic fields are chiral/anti-chiral

$$\psi = \psi(z)$$

$$\bar{\psi} = \bar{\psi}(\bar{z})$$

* Determine the conformal properties of ψ and $\bar{\psi}$

if take $w = \lambda z$, the action should be invariant

$$(i) S = \frac{1}{4\pi\alpha'} \int d^2z (\psi'(z, \bar{z}) \bar{\partial}_z \psi'(z, \bar{z}) + \bar{\psi}'(z, \bar{z}) \partial_{\bar{z}} \bar{\psi}'(z, \bar{z}))$$

$$= \frac{1}{4\pi\alpha'} \int \frac{dz}{\partial w} d\omega \frac{d\bar{z}}{\partial \bar{w}} d\bar{\omega} [\psi'(z, \bar{z}) (\frac{\partial \bar{z}}{\partial \bar{w}}) \bar{\partial}_{\bar{w}} \psi'(z, \bar{z})$$

$$+ \dots]$$

$$= \frac{1}{4\pi\alpha'} \int d\omega d\bar{\omega} \left(\frac{dz}{d\omega} \right) \psi'(\omega, \bar{\omega}) \bar{\partial}_{\bar{\omega}} \psi'(\omega, \bar{\omega}) + \dots$$

$$= \frac{1}{4\pi\alpha'} \int d\omega d\bar{\omega} [\psi(\omega, \bar{\omega}) \bar{\partial}_{\bar{\omega}} \psi(\omega, \bar{\omega}) + \dots]$$

$$\therefore \left(\frac{\partial \bar{z}}{\partial \bar{w}} \right) \psi'(z, \bar{z}) \bar{\partial}_{\bar{w}} \psi'(z, \bar{z}) = \psi(\omega, \bar{\omega}) \bar{\partial}_{\bar{\omega}} \psi(\omega, \bar{\omega})$$

$$\text{so } \left[\left(\frac{\partial \bar{z}}{\partial \bar{w}} \right)^{1/2} \psi'(z, \bar{z}) \right] \bar{\partial}_{\bar{\omega}} \left[\left(\frac{\partial \bar{z}}{\partial \bar{w}} \right)^{1/2} \psi'(z, \bar{z}) \right]$$

$$\text{or } \psi(\omega, \bar{\omega}) = \left(\frac{dz}{d\omega} \right)^{1/2} \psi(z, \bar{z})$$

$$\psi: h = 1/2, \bar{h} = 0 \quad | \quad (h, \bar{h}) = (1/2, 0)$$

$$\bar{\psi}: h = 0, \bar{h} = 1/2 \quad | \quad (h, \bar{h}) = (0, 1/2)$$

$\Rightarrow$ Has different behavior under 2π rotation

$$\psi(e^{2\pi i} z) = +\psi(z)$$

$$\bar{\psi}(e^{2\pi i} z) = -\bar{\psi}(z)$$

Neveu-Schwartz sector (NS)

Ramond sector (R)

Radial ordering:

$$R(\psi(z) \phi(w)) := \begin{cases} +\psi(z) \phi(w) & \text{for } |z| > |w| \\ -\phi(w) \psi(z) & \text{for } |z| < |w| \end{cases}$$

$$\psi(z) = \sum_r \psi_r z^{-r-1/2}$$

However, due to NS and R conditions the values or differ

(NS): $\psi(z) = + \psi(e^{2\pi i} z)$

$$\sum_r z^{-r-1/2} \psi_r = \sum_r e^{-2\pi i(r+1/2)} z^{-r-1/2} \psi_r$$

$$e^{-2\pi i r} e^{-i\pi} = -e^{-2\pi i r} \stackrel{!}{=} 1$$

$$\therefore -2\pi i r = -(2n+1)\pi i$$

$$\therefore r = (n + 1/2)$$

$$r \in \mathbb{Z} + 1/2$$

(R): $\psi(e^{2\pi i} z) = -\psi(z)$

$$\therefore e^{-2\pi i(r+1/2)} = -e^{-2\pi i r}$$

$$r \in \mathbb{Z}$$

$$\Rightarrow \begin{cases} \psi_{NS}(z) = \sum_r \psi_r z^{-r-1/2}, & r \in \mathbb{Z} + 1/2 \\ \psi_R(z) = \sum_r \psi_r z^{-r-1/2}, & r \in \mathbb{Z} \end{cases}$$

on $S^1 \times \mathbb{R}$ the NS, R conditions are

$$\psi_+^u(\pi) = \psi_-^u(\pi) \quad (R)$$

$$\psi_+^u(\pi) = -\psi_-^u(\pi) \quad (NS)$$

$$G(z, w) = \langle \psi(z) \psi(w) \rangle$$

$$\int D\psi D\bar{\psi} \frac{\delta}{\delta \psi(z, \bar{z})} \int \exp(-S[\psi, \bar{\psi}]) \psi(z)$$

$$\left(\langle \psi(z) \psi(w) \rangle = \int D\psi e^{-S[\psi]} \psi(z) \psi(w) \right.$$

$$\rightarrow \int D\psi \left(\psi(z) + \delta^2(z-w) e^{\delta S[\psi]} \right)$$

$$\left((-\delta S / \delta \psi(w, \bar{w})) \psi(w) + \delta^2(z-w) \right)$$

$$\int D\psi e^{(-2\pi \int d\omega \psi(\omega) \partial \psi(\omega) \psi(z) + \partial(z-\omega))}$$

$$\therefore \text{we normalize } \int D\psi e^{-S} = 1$$

$$\therefore \bar{\partial} \langle \psi(\omega) \psi(z) \rangle = -2\pi \delta^2(z-\omega)$$

• However, we note

$$\bar{\partial} z^{-1} = 2\pi \delta^2(z)$$

$$\therefore \begin{aligned} \langle \psi(z) \psi(\omega) \rangle &= -\frac{1}{(z-\omega)} \\ \langle \bar{\psi}(\bar{z}) \bar{\psi}(\bar{\omega}) \rangle &= -1/(\bar{z}-\bar{\omega}) \end{aligned}$$

→ Note: the Green's fn satisfies the Grassmann odd properties of $z \leftrightarrow \omega$ which is invariant.

Wicks thm: $\psi(z)\psi(\omega) = \langle \psi(z)\psi(\omega) \rangle + : \psi(z)\psi(\omega) :$
 $= -1/(z-\omega) + \dots$

$$\bar{\psi}(\bar{z})\bar{\psi}(\bar{\omega}) = -\frac{1}{(\bar{z}-\bar{\omega})} + \dots, \quad \bar{\psi}(\bar{z})\psi(\omega) = 0 + \dots$$

E.M. tensor: Use the canonical energy-momentum tensor for a theory w fields ϕ_i and Lagrangian $\mathcal{L}$.

$$T_{\mu\nu}^c = 8\pi\gamma \left(-\eta_{\mu\nu} \mathcal{L} + \sum_i \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_i)} \partial_\nu \phi_i \right) \quad \left(\begin{array}{c} \text{Canonical} \\ \text{E.M. tensor} \end{array} \right)$$

$$\text{use } g_{ab} = \begin{pmatrix} 0 & 1/2 \\ 1/2 & 0 \end{pmatrix}$$

$$\text{Note } \partial^z = 2\bar{\partial}_{\bar{z}}, \quad \partial^{\bar{z}} = 2\partial_z$$

$$\frac{\delta \mathcal{L}}{\delta(\partial_\mu \psi)} \partial_\nu \psi + \frac{\delta \mathcal{L}}{\delta(\partial_\mu \bar{\psi})} \partial_\nu \bar{\psi}$$

$$T_{zz} = \frac{1}{2} \left(\frac{\delta \mathcal{L}}{\delta(\partial_z \psi)} \partial_z \psi + \frac{\delta \mathcal{L}}{\delta(\partial_z \bar{\psi})} \partial_z \bar{\psi} \right) = \frac{1}{2} \psi \partial \psi$$

$$T_{z\bar{z}} = 0 \quad \leftarrow \text{From E.O.M } \bar{\partial} \psi = \partial \bar{\psi} = 0$$

$$T_{\bar{z}z} = 0$$

$$\bar{T}_{\bar{z}\bar{z}} = \frac{1}{2} \bar{\psi} \bar{\partial} \bar{\psi}$$

$$\therefore T = \frac{1}{2} : \psi \partial \psi : , \quad \bar{T} = \frac{1}{2} : \bar{\psi} \bar{\partial} \bar{\psi} :$$

$$\begin{aligned} \sim \mathcal{D} T(z) \psi(w) &= \frac{1}{2} : \psi(z) \partial_z \psi(z) : \psi(w) \\ &= \left(\frac{1}{2} \psi(z) \partial_z \psi(z) \psi(w) \right) \\ &\quad - \frac{1}{2} \langle \psi(z) \psi(w) \rangle \partial_z \psi(z) + \frac{1}{2} \psi(z) \langle \partial_z \psi(z) \psi(w) \rangle \\ &= (\text{Reg}) - \frac{1}{2} \left(-\frac{1}{z-w} \right) \partial_z \psi(z) + \frac{1}{2} \psi(z) \frac{1}{(z-w)^2} \\ &= (\text{Reg}) + \frac{1}{2} \frac{1}{(z-w)} \sum_{n=0}^{\infty} \frac{(z-w)^n}{n!} \partial^n \psi(w) + \frac{1}{2} \frac{1}{(z-w)^2} \sum_n \frac{(z-w)^n}{n!} \partial^n \psi(w) \end{aligned}$$

$$\begin{aligned} (1^{st}) &= \frac{\partial \psi(w)}{(z-w)} + \text{Reg} \\ (2^{nd}) &= \frac{(1/2) \psi(w)}{(z-w)^2} + \frac{1}{2} \frac{\partial \psi(w)}{(z-w)} \end{aligned}$$

$$\therefore T(z) \psi(w) = \frac{(1/2) \psi(w)}{(z-w)^2} + \frac{3/2 \partial \psi(w)}{(z-w)} + \dots$$

and so we further confirm ψ has conformal weight $h = 3/2$. and $\bar{h} = 3/2$ for $\bar{\psi}$.

* Algebra of Laurent modes:

$$\psi_r = \oint \frac{dz}{2\pi i} \psi(z) z^{r-1/2} (-i)$$

$$\{\psi_r, \psi_s\} = \oint \frac{dz}{2\pi i} \oint \frac{dw}{2\pi i} \{\psi(z), \psi(w)\} z^{r-1/2} w^{s-1/2}$$

$$\oint \frac{dz}{2\pi i} = \oint_{|z| > |w|} \frac{dz}{2\pi i} - \oint_{|z| < |w|} \frac{dz}{2\pi i}$$

$$= \oint \frac{dw}{2\pi i} \left[\oint_{|z| > |w|} \frac{dz}{2\pi i} \psi(z) \psi(w) - \oint_{|z| < |w|} \frac{dz}{2\pi i} (-\psi(w) \psi(z)) \right] z^{r-1/2} w^{s-1/2}$$

$$= \oint \frac{dw}{2\pi i} \oint \frac{dz}{2\pi i} R(\psi(z) \psi(w)) z^{r-1/2} w^{s-1/2}$$

$$= \oint \frac{dw}{2\pi i} \oint \frac{dz}{2\pi i} \left(\frac{1}{z-w} \right) z^{r-1/2} w^{s-1/2}$$

$$\oint \frac{dw}{2\pi i} w^{r+s-1} \Rightarrow r+s$$

$$\sim \oint \frac{dw}{2\pi i} w^{r+s-1} = \delta_{r+s, -1}$$

$$\{\psi_r, \psi_s\} = \delta_{r+s,0} \quad | \quad \{b_r, b_s\} = \delta_{r+s,0} \quad | \quad \{b_r, b_s\} = 0$$

West uses: $\psi = -i \sum_r b_r z^{-r-1/2}$

$\Rightarrow$ By hermitian properties b_r : $b_r^\dagger = b_{-r}$

Vacuum is defined by: $r \in \mathbb{Z} + 1/2$

$$\lim_{z \rightarrow 0} (\psi(z) |0\rangle) = \lim_{z \rightarrow 0} \left(\sum_r z^{-r-1/2} (b_r |0\rangle) \right)$$

$$\underline{r=0}: \lim_{z \rightarrow 0} z^{-1/2} b_0 |0\rangle$$

$\therefore$

$$b_r |0\rangle = 0 \quad \forall r \geq 0$$

Normal ordering:

$$:b_r b_s: = \begin{cases} b_r b_s, & s > r \\ -b_s b_r, & r > s \end{cases}$$

• T has the same conf. dim regardless of our theory:

$$T(z) = \sum_n L_n z^{-n-2} = \frac{1}{2} : \psi(z) \partial \psi(z) :$$

(RHS)

$$= -\frac{1}{2} : \sum_r z^{-r-1/2} \psi_r \sum_s (-s-1/2) z^{-s-3/2} \psi_s$$

$$= \frac{1}{2} : \sum_{r,s} z^{-r-s-2} (s+1/2) : \psi_r \psi_s :$$

$$= \frac{1}{2} \sum_n z^{-n-2} \sum_s' (s+1/2) : \psi_{n-s} \psi_s :$$

$s+r=n, \quad r=n-s$

$$L_n = \frac{1}{2} \sum_s (s+1/2) : \psi_{n-s} \psi_s :$$

Using West's def:

$$= \frac{1}{2} \sum_{s > -3/2} \psi_{n-s} \psi_s (s+1/2) - \frac{1}{2} \sum_{s \leq -3/2} \psi_s \psi_{n-s} (s+1/2)$$

• Central charge: $\langle 0 | L_2 L_{-2} | 0 \rangle = c/2$

$$L_2 = \frac{1}{2} \sum \psi_{-n} \psi_n (n+1/2) = \frac{1}{2} \sum \psi_{-n} \psi_n (n+1/2)$$

$$= -\frac{1}{4} \psi_{-1} \psi_{-1} + \frac{1}{2} \psi_{-3/2} \psi_{-1/2}$$

Not sure why this term isn't present.

$$\langle 0 | L_2, L_{-2} | 0 \rangle = 1/4$$

$$\therefore C_\psi = \pm 1/2$$

Complex fermions & Bosonification:

* In 2D it is possible to express fermions in terms of Bosons and vice-versa. A free scalar w/ no background charge has central charge $(1, 1)$, for $\psi_L(z)$ and $\psi_R(z)$

~> A single Majorana fermion consists of the real components ψ and $\bar{\psi}$, each w/ conformal weights $(\pm 1/2, 0)$, $(0, \pm 1/2)$

Requirement: Each respective E.M. tensor have the same left and right central charges.

Idea: $\phi : (1, 1) = (c, \bar{c}) \longrightarrow (\psi^{(1)} \oplus \psi^{(2)}) \oplus (\bar{\psi}^{(1)} \oplus \bar{\psi}^{(2)}) = (1, 1) = (c, \bar{c})$

Take two left handed weyl spinors of $\psi^{(1)}, \psi^{(2)}$ &

$$S = -\frac{1}{4\pi} \int d^2z \, d\bar{z} \{ \psi^{(1)} \bar{\partial} \psi^{(1)} + \psi^{(2)} \bar{\partial} \psi^{(2)} \}$$

then the Maj.

* Note: we assumed: $\psi^\dagger = \psi$ $\bar{\psi}^\dagger = \bar{\psi}$

Since we work with the Maj. field

$$\Psi = \begin{pmatrix} \psi \\ \bar{\psi} \end{pmatrix} : \quad \bar{\Psi}^c = \bar{\Psi} \quad \xrightarrow{\text{implies}}$$

$$S = -\frac{1}{4\pi} \int d^2z d\bar{z} \{ \psi^{(1)} \bar{\partial} \psi^{(1)} + \psi^{(2)} \bar{\partial} \psi^{(2)} \} = -\frac{1}{4\pi} \int d^2z d\bar{z} \{ \Psi^* \bar{\partial} \Psi + \Psi \bar{\partial} \Psi^* \}$$

$$\Psi^\pm \equiv \frac{1}{\sqrt{2}} (\psi^{(1)} \pm i \psi^{(2)})$$

Since $\Psi^\pm$ is a sum of two spin 1/2 fields:

$$\Psi^\pm = \sum_r \Psi_r^\pm z^{-r-1/2}$$

with

$$\{ \Psi_r^+, \Psi_s^+ \} = 0, \{ \Psi_r^+, \Psi_s^- \} = \delta_{r+s,0}$$

$\Rightarrow$ The E.M. tensor: (Chiral)

$$T = \frac{1}{2} : \psi^{(1)} \partial \psi^{(1)} : + \frac{1}{2} : \psi^{(2)} \partial \psi^{(2)} : \\ = \frac{1}{2} : \Psi^+ \partial \Psi^+ : + \frac{1}{2} : \Psi^- \partial \Psi^- :$$

Proof: $\frac{1}{\sqrt{2}} (\Psi^+ + \Psi^-) = \psi^{(1)}, \quad \psi^{(2)} = \frac{1}{\sqrt{2}} i (\Psi^+ - \Psi^-)$

$$\begin{aligned} \psi^{(1)} \partial \psi^{(1)} &= \frac{1}{2} (\Psi^+ + \Psi^-) \partial (\Psi^+ + \Psi^-) \\ \psi^{(2)} \partial \psi^{(2)} &= -\frac{1}{2} (\Psi^+ - \Psi^-) \partial (\Psi^+ - \Psi^-) \\ &= \frac{1}{2} [\Psi^+ \partial \Psi^+ + \Psi^+ \partial \Psi^- + \Psi^- \partial \Psi^+ + \Psi^- \partial \Psi^- \\ &\quad - \Psi^+ \partial \Psi^- - \Psi^- \partial \Psi^+] \\ &= \frac{1}{2} [\Psi^+ \partial \Psi^+ + \Psi^- \partial \Psi^-] \\ &\text{as claimed} \end{aligned}$$

*Relating $\Psi^\pm$ and the scalar field $\varphi_L(z)$:

$$\begin{aligned} \Psi^+(z) &= c_+ : e^{i\varphi_L(z)} : \quad \hookrightarrow h = 1/2 \\ \Psi^-(z) &= c_- : e^{-i\varphi_L(z)} : \quad \hookrightarrow h = 1/2 \end{aligned}$$

$\Rightarrow$ Since $: e^{ik_1 \varphi_L(z)} : : e^{ik_2 \varphi_L(w)} :$

$$= (z-w)^{k_1 k_2} : e^{i k_1 \varphi_L + i k_2 \varphi_L(w)} :$$

$$\Psi^+(z) \Psi^+(w) = (c_+)^2 (z-w)^{-1} e^{i(\varphi_L(z) + \varphi_L(w))} = 0 + \dots$$

$$\Psi^+(z) \Psi^-(w) = (c_+)^2 (z-w) e^{i(\phi_L(z) + \phi_L(w))} = 0$$

which vanish in the limit $z \rightarrow w$ and these are part of $\phi(z)$: i.e. the "regular" part of the OPE

$$\Psi^+(z) \Psi^-(w) = c_+ c_- (z-w)^{-1} e^{i(\phi_L(z) - \phi_L(w))} + \dots$$

$$\begin{aligned} \Psi^+(z) \Psi^-(w) &= \frac{1}{2} (\psi^{(1)}(z) + i \psi^{(2)}(z)) (\psi_L^{(1)}(w) - i \psi_L^{(2)}(w)) \\ &= \frac{1}{2} (\psi^{(1)}(z) \psi_L^{(1)}(w) + \psi^{(2)}(z) \psi_L^{(2)}(w)) \\ &= \frac{1}{2} \left(-\frac{1}{z-w} - \frac{i}{z-w} \right) \\ &\sim -\frac{1}{z-w} \end{aligned}$$

exactly that of a single free ferm. field:

$$\boxed{c_+ c_- = -1}$$

$$\begin{aligned} \Psi^+(z) \Psi^-(w) &= \frac{-1}{z-w} e^{i(\phi_L(z) - \phi_L(w))} \\ &= (-1) \sum_{n=0}^{\infty} \frac{(i(\phi_L(z) - \phi_L(w)))^n}{n!} \end{aligned}$$

$$\phi_L(z) - \phi_L(w) = \sum_{l=0}^{\infty} \frac{(z-w)^l}{l!} \partial^l \phi_L(w) - \phi_L(w)$$

$$\begin{aligned} &= \phi_L(w) - \phi_L(w) + (z-w) \partial \phi_L(w) + \mathcal{O}((z-w)^2) \\ &= (z-w) \partial \phi_L(w) + \mathcal{O}((z-w)^2) \end{aligned}$$

$$\begin{aligned} \Psi^+(z) \Psi^-(w) &= \frac{-1}{z-w} (1 + i(z-w) \partial \phi_L(w) + \mathcal{O}((z-w)^2)) \\ &= \frac{-1}{z-w} - i \partial \phi_L(w) + \mathcal{O}(z-w) \end{aligned}$$

$$\boxed{:\Psi^+(z) \Psi^-(w): = i \partial \phi_L(w)} \quad (\text{diff by a } -1)$$

→ Differentiating the above wrt. w :

$$:\Psi^+(z) \partial \Psi^-(z): = -c_+ c_- \left(i \frac{(\partial \phi_L)^2}{2} + \frac{i}{2} \partial^2 \phi_L \right)$$

Adding $:\Psi^\dagger(z)\partial\Psi^\dagger(z):$ we find that
 $T = -\frac{1}{2}:(\partial\varphi_L)^2/2:$

Let's do this:

$$\begin{aligned}\Psi^\dagger(z)\partial_w\Psi^\dagger(w) &= \frac{C_+C_-}{(z-w)^2}e^{i(P_Lz-\varphi_Lw)} + \frac{C_+C_-}{(z-w)}\frac{-i\partial\varphi_L(w)}{(z-w)} \\ &= \left(\frac{C_+C_-}{(z-w)^2} - \frac{i\partial\varphi_L(w)}{(z-w)}\right)\left(1 + (z-w)\partial\varphi_L(w) + (z-w)^2\partial^2\varphi_L(w)\right) \\ &= \left(\frac{C_+C_-}{(z-w)^2} - \frac{i\partial\varphi_L(w)}{(z-w)}\right)\left(1 + i(z-w)\partial\varphi_L(w) + \frac{1}{2}i^2(z-w)^2\partial^2\varphi_L(w) + \frac{1}{2}i^2(z-w)^2(\partial\varphi_L(w))^2 + \mathcal{O}((z-w)^3)\right) \\ &= (\text{Singular}) + \partial\varphi_L(w)\partial\varphi_L(w) + iC_+C_-\partial^2\varphi_L(w)\frac{1}{2} - \frac{1}{2}C_+C_-(\partial\varphi_L(w))^2\end{aligned}$$

First order Systems: two Grassmann odd fields (b, c) with conf. weights

$$b: (\lambda, 0)$$

$$c: (1-\lambda, 0)$$

$\Rightarrow$ Conf. transf.

$$\delta b = \epsilon\partial b + \lambda(\partial\epsilon)b + \bar{\epsilon}\bar{\partial}b$$

$$\delta c = \epsilon\partial c + (1-\lambda)(\partial\epsilon)c + \bar{\epsilon}\bar{\partial}c$$

Action (in $\mathbb{H}^2$):

$$S = \frac{1}{2\pi}\int dz d\bar{z} b\bar{\partial}c$$

• Since

$$S = \frac{1}{2\pi}\int d^2x b_{mn}\partial^m c^n$$

b_{mn} : symmetric traceless

$$g^{\mu\nu}b_{\mu\nu} = 0 \quad b_z\bar{z} + b_{\bar{z}}z = 0$$

$$b_{zz} \equiv b(z) \quad | \quad b_{\bar{z}\bar{z}} \equiv \bar{b}(\bar{z})$$

$$c^z \equiv c(z) \quad | \quad \bar{c}^{\bar{z}} \equiv \bar{c}(\bar{z})$$

$$\Rightarrow \delta S^{b,c} = \frac{1}{2\pi} \int dz d\bar{z} \delta (b \bar{\partial} c)$$

$$= \frac{1}{2\pi} \int dz d\bar{z} \left(\epsilon \partial b + \lambda (\partial \epsilon) b + \bar{\epsilon} \bar{\partial} b \right) \bar{\partial} c$$

$$+ b \bar{\partial} \left(\epsilon \partial c + (1-\lambda) (\partial \epsilon) c + \bar{\epsilon} \bar{\partial} c \right)$$

$$= \frac{1}{2\pi} \int dz d\bar{z} \left(\bar{\partial} \epsilon \right) \left(\lambda b \partial c - (1-\lambda) \partial b c \right)$$

Proof:

$$\int \left(\epsilon \partial b \bar{\partial} c + \lambda (\partial \epsilon) b \bar{\partial} c + \bar{\epsilon} \bar{\partial} b \bar{\partial} c \right.$$

$$+ b (\bar{\partial} \epsilon) \partial c + b \epsilon \bar{\partial} \partial c + b (1-\lambda) (\bar{\partial} \partial \epsilon) c + b (1-\lambda) \partial \bar{\partial} \epsilon c$$

$$+ b \bar{\partial} \bar{\epsilon} \bar{\partial} c + b \bar{\epsilon} \bar{\partial}^2 c \Big)$$

ϵ terms: $\epsilon \partial b \bar{\partial} c + \lambda (\partial \epsilon) b \bar{\partial} c + b \epsilon \bar{\partial} \bar{\partial} c$

$$+ b \bar{\partial} \partial \epsilon c - \lambda b \bar{\partial} \partial \epsilon c + b (\partial \epsilon) \bar{\partial} c$$

$$- \lambda b \partial \epsilon \bar{\partial} c$$

- $\epsilon \partial b \bar{\partial} c = - b \bar{\partial} \epsilon \bar{\partial} c - \epsilon b \bar{\partial} \partial c$

no left with

$$\lambda (\partial \epsilon) b \bar{\partial} c + b \bar{\partial} \partial \epsilon c - \lambda b \bar{\partial} \partial \epsilon c - \lambda b \partial \epsilon \bar{\partial} c$$

$$= b c \left(\bar{\partial} \partial \epsilon \right) (1-\lambda)$$

$\bar{\epsilon}$ terms: $\bar{\epsilon} \bar{\partial} b \bar{\partial} c + b \bar{\partial} \bar{\epsilon} \bar{\partial} c + b \bar{\epsilon} \bar{\partial}^2 c$

* Should be: $g_{ab} = \begin{pmatrix} 0 & 1/2 \\ 1/2 & 0 \end{pmatrix}, g^{ab} = \begin{pmatrix} 0 & 2 \\ 2 & 0 \end{pmatrix}$

$$\frac{1}{4\pi} \int dz d\bar{z} \left(b_{zz} \partial^z c^z + b_{\bar{z}\bar{z}} \partial^{\bar{z}} c^{\bar{z}} \right)$$

$$b_{zz} \equiv b(z) \quad c^z \equiv c(z)$$

$$b_{\bar{z}\bar{z}} \equiv \bar{b}(\bar{z}) \quad c^{\bar{z}} \equiv \bar{c}(\bar{z})$$

$$\partial^z = 2 \bar{\partial}_{\bar{z}}, \quad \partial^{\bar{z}} = 2 \partial_z$$

$$\frac{1}{2\pi} \int dz d\bar{z} \left(b \bar{\partial} c + \bar{b} \partial \bar{c} \right)$$

• Canonical E.M. tensor is

$$T_{\mu\nu}^c = 8\pi\kappa \left(-\eta_{\mu\nu} \mathcal{L} + \sum_i \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi_i)} \partial_\nu \phi_i \right)$$

and also only look at $\bar{E}$ terms and

✓ went wrong somewhere. (9.2.3) is correct

$$\delta S^{b,c} = \frac{1}{2\pi} \int dz d\bar{z} (\bar{\partial} E) (\lambda b \partial c - (1-\lambda) \partial b c)$$

• For a conf. trans. $\bar{\partial} E = 0$

$$T_{z\bar{z}}^{b,c} = - : \lambda b \partial c + (1-\lambda) \partial b c :$$

• Conclude $\bar{T}^{b,c} = 0$

* Expression for $(0, \lambda)$, $(0, 1-\lambda)$ are same

E.O.M: $\bar{\partial} b = 0 = \bar{\partial} c$

$$\int D b D c \frac{\delta}{\delta b(w)} (e^{-S[b,c]} b(z)) = 0$$

$$\int D b D c \left(-\frac{\delta S}{\delta b} b(z) \right) + \delta(z-w) e^{-S[b,c]} = 0$$

$$\frac{\delta S}{\delta b(w)} = \frac{1}{2\pi} \bar{\partial} c(w)$$

$$\int D b D c \left(e^{-S[b,c]} \left(\frac{\bar{\partial} c(w)}{2\pi} b(z) \right) \right) = \delta^2(z-w) 1$$

$$\langle \bar{\partial} c(w) b(z) \rangle = 2\pi \delta^2(z-w)$$

$$= \bar{\partial}_w \langle c(w) b(z) \rangle = 2\pi \delta(z-w)$$

$$\bar{\partial}_w ((w-z)^{-1}) = 2\pi \delta(z-w)$$

$$\langle c(w) b(z) \rangle = z/(w-z)$$

$$-\langle b(z) c(w) \rangle = z/(w-z)$$

and we conclude

$$\langle b(z) c(w) \rangle = \frac{1}{(z-w)}$$

$$* R(c(z) b(w)) = : c(z) b(w) : + \langle c(z) b(w) \rangle$$

$$OPE: c(z) b(w) = \frac{1}{z-w} + \dots$$

$$b(z) c(w) = \frac{1}{z-w} + \dots$$

* c, b are meromorphic fvs from their F.O.M

$$\Rightarrow c(z) = \sum_{-\infty}^{\infty} c_n z^{-n-(1-\lambda)}$$

$$b(z) = \sum_{-\infty}^{\infty} b_n z^{-n-\lambda}$$

* The Vacuum $|0\rangle$ is defined to satisfy

$$\bullet \lim_{z \rightarrow 0} \sum_{-\infty}^{\infty} z^{-n-(1-\lambda)} c_n |0\rangle$$

$$n + (1-\lambda) \geq 1$$

$$\Rightarrow c_n |0\rangle = 0$$

$$\boxed{n \geq \lambda}$$

$$c_n |0\rangle = 0$$

$$\bullet \lim_{z \rightarrow 0} \sum_{-\infty}^{\infty} b_n z^{-n-\lambda} |0\rangle$$

$$n + \lambda \geq 1 \Rightarrow$$

$$\boxed{n \geq 1 - \lambda}$$

$$b_n |0\rangle$$

The E.M. tensor is of the form

$$T(z) = \sum L_m^{b,c} z^{-m-2} \quad | \quad L_m^{b,c} = \oint \frac{dz}{2\pi i} z^{m+1} T^{b,c}(z)$$

$$L_n = \oint \frac{dz}{2\pi i} z^{n+1} \left(-\lambda b \partial c + (1-\lambda) \partial b c \right)$$

$$|i\rangle + \oint \frac{dz}{2\pi i} z^{n+1} \sum b_n z^{-n-\lambda} \sum c_n z^{-n-(1-\lambda)-1}$$

$$= + \oint \frac{dz}{2\pi i} \lambda \sum_{m, l} : b_m c_l : (l + (1-\lambda))$$

$$n+1 - m - \lambda - l - 1 + \lambda - 1 = -1$$

$$n - m - l = 0$$

$$m = n - l$$

$$= \lambda \sum_l (l + (1-\lambda)) : b_{n-l} c_l :$$

$$(ii) + (1-\lambda) \oint \frac{dz}{2\pi i} z^{n+1} \sum_{m, l} : b_m c_l : (m + \lambda) z^{-m-\lambda-1} z^{-l-(1-\lambda)}$$

$$n+1 - m - \lambda - l - 1 + \lambda = -1$$

$$m = n - l$$

$$= (1-\lambda) \sum_l : b_{n-l} c_l : (n - l + \lambda)$$

$$= \sum_l : b_{n-l} c_l : \left[\lambda(l + (1-\lambda)) + (1-\lambda)(n - l + \lambda) \right]$$

$$\lambda l + \lambda - \lambda^2 = n + n\lambda - l - l\lambda + \lambda + \lambda^2$$

$$= (n(\lambda - 1) - l)$$

$$\therefore L_n = \sum_p : b_{n-p} c_p : (n(\lambda - 1) - p)$$

and West has

$$(p - (1-\lambda)n) = (p + (\lambda - 1)n)$$

so: off by two minus signs

one for $n \rightarrow -n$ and $p \rightarrow -p$

* The vacuum is $SL(2, \mathbb{C})$ invariant:

$$L_0 = \sum_p : b_{-p} c_p : (p)$$

$$\approx \sum_{p < 0} (-1)^p C_p b_{-p} |p\rangle + \sum_{p \geq 0} b_{-p} C_p |p\rangle$$

Suppose $\lambda \geq 0$:

$$= \left(\sum_{p=\lambda}^{\infty} b_{-p} C_p |p\rangle + \sum_{p=0}^{\lambda-1} b_{-p} C_p |p\rangle + \sum_{p=-\infty}^{-1} C_p b_{-p} |p\rangle \right) |0\rangle$$

First term vanishes since $C_p |0\rangle = 0$ for $p \geq \lambda$

$$\bullet \sum_{p=0}^{\lambda-1} b_{-p} C_p |p\rangle + \sum_{p=-\infty}^{-1} C_p b_{-p} |p\rangle$$

$$= \left(\sum_{p=-\infty}^{\lambda-1} + \sum_{p=\lambda}^{\infty} \right) : b_{-p} C_p : |p\rangle$$

$$= : b_{\lambda-1} C_{1-\lambda} : (1-\lambda) + : b_1 C_{-\lambda} : (-1) + (-\lambda-1) : b_{\lambda+1} C_{-\lambda-1} :$$

So first term vanishes for all but

$$-1 + \lambda \geq -1 \Rightarrow 1 - \lambda \leq 1$$

trivial if $\lambda = 0$:

$\lambda = 1$: first term vanishes and so does

sec.

$$\lambda \geq 1: \sum_{p=-\infty}^{\lambda-1} \text{vanishes:}$$

$$\sum_{p=\lambda}^{\infty} : b_{-p} C_p : |p\rangle |0\rangle$$

$$b_{1-\lambda} C_{\lambda-1} (k-1) \quad \lambda-1$$

$\lambda > 1$:

$\lambda = 2$:

Suppose $\lambda = z$ think this should vanish...

* Con. show: $\{c_n, b_m\} = \delta_{n+m, 0}$

Proof:

$$\begin{aligned} \{c_n, b_m\} &= \oint_{2\pi i} \frac{dz}{z} z^{n-1+\lambda} \oint_{2\pi i} \frac{dw}{z} w^{m-1+(1-\lambda)} \mathcal{R}((z-w)^{-1}) \\ &= \oint_{2\pi i} \frac{dz}{z} \oint_{2\pi i} \frac{dw}{z} z^{n-1+\lambda} w^{m-1+(1-\lambda)} \left(\frac{1}{z-w} \right) \\ &= - \oint_{2\pi i} \frac{dz}{z} z^{-n-1+\lambda+m-\lambda} \\ &= \int_{-1}^{n-1+m} z^{-1} = \delta_{n+m, 0} \\ &= \delta_{n+m, 0} \end{aligned}$$

off by a (-) sign

$\therefore \{c_n, b_m\} = \delta_{n+m, 0} \mid \{c_n, c_m\} = 0 = \{b_n, b_m\}$

OPE w $T^{b,c}(z)$:

$T^{b,c}(z) b(w) = (-\lambda : \partial b c : + (1-\lambda) b \partial c) b(w)$

$T^{b,c}(z) b(w) = \frac{\lambda b(w)}{(z-w)^2} + \frac{\partial b(w)}{(z-w)} + \dots$

$T^{b,c}(z) c(w) = -\lambda : b \partial c : c(w) + (1-\lambda) : \partial b c : c(w)$

$$\begin{aligned} &= (\text{reg.}) + \lambda \partial c(z) \langle b(z) c(w) \rangle \\ &+ (1-\lambda) \partial_z \langle b(z) c(w) \rangle c(z) \\ &= \lambda \partial c(z) \frac{1}{z-w} + (1-\lambda) \frac{1}{(z-w)^2} c(z) \\ &= \lambda \partial c(w) \frac{1}{z-w} + \frac{(1-\lambda)}{(z-w)^2} [c(w) + (z-w) \partial c(w)] \\ &+ \dots \\ &= \lambda \partial c(w) \frac{1}{(z-w)} + \frac{(1-\lambda) c(w)}{(z-w)^2} + (1-\lambda) \frac{\partial c(w)}{(z-w)} + \dots \end{aligned}$$

$T^{b,c}(z) c(w) = (1-\lambda) c(w) / (z-w)^2 + \frac{\partial c(w)}{(z-w)} + \dots$

$$T^{b,c}(z) + b^c(w) = \frac{1}{2} \frac{1}{(z-w)^4} + \frac{1}{(z-w)^2} + \frac{1}{z-w} \partial_w T^{b,c}(w) + \dots$$

$$\text{with } c^{b,c} = 2(-1 + 6\lambda - 6\lambda^2)$$

$$\text{proof: } (-\lambda : b \partial c : + (1-\lambda) : \partial b c :)(-\lambda : b \partial c : + (1-\lambda) : \partial b c :)$$

$$(i) \lambda^2 : b(z) \partial c(z) : : b(w) \partial c(w) :$$

$$= \lambda^2 : b(z) \partial c(w) : \langle \partial c(z) b(w) \rangle + \langle b(z) \partial c(w) \rangle : \partial c(z) :$$

$$+ \langle b(z) \partial c(w) \rangle \langle \partial c(z) b(w) \rangle \frac{1}{z-w}$$

$$= \lambda^2 \left(: b(z) \partial c(w) : \left(-\frac{1}{(z-w)^2} \right) + \lambda^2 \frac{1}{(z-w)^2} : \partial c(z) b(w) : \right)$$

$$= \lambda^2 \frac{1}{(z-w)^2} \left(\frac{1}{(z-w)^2} \right)$$

$$= \frac{\lambda^2}{(z-w)^2} \left[- : b(z) \partial c(w) : + : \partial c(z) b(w) : \right]$$

$$= \lambda^2 / (z-w)^4$$

$$: b(z) \partial c(w) : = : b(w) \partial c(w) : + : (z-w) \partial b \partial c :$$

$$: \partial c(z) b(w) : = : \partial c(w) b(w) : + (z-w) : \partial^2 c(w) b(w) :$$

$$(i) = \frac{\lambda^2}{(z-w)^2} \left[- : b(w) \partial c(w) : + : \partial c(z) b(w) : \right]$$

$$+ \lambda^2 / (z-w) \left[- : \partial b(w) c(w) : + : \partial^2 c(w) b(w) : \right]$$

$$= \lambda^2 / (z-w)^4$$

$$(ii) \lambda(\lambda-1) : b(z) \partial c(z) : = \partial b(w) c(w) :$$

$$= \text{reg} + \lambda(\lambda-1) : b(z) c(w) : \langle \partial c(z) \partial b(w) \rangle$$

$$+ \lambda(\lambda-1) \langle b(z) c(w) \rangle : \partial c(z) \partial b(w) :$$

$$+ \lambda(\lambda-1) \langle b(z) c(w) \rangle \langle \partial c(z) \partial b(w) \rangle$$

$$= \lambda(\lambda-1) \left(\frac{1}{(z-w)^3} : b(z) c(w) : + : \partial c(z) \partial b(w) : \frac{1}{z-w} \right)$$

$$+ \lambda(\lambda-1) \left(\frac{1}{(z-w)} \left(\frac{-2}{(z-w)^3} \right) \right)$$

$$= \lambda(\lambda-1) \left(\frac{-2}{(z-w)^3} \right) \left(: b(w) c(w) : + (z-w) : \partial b c(w) : + \frac{1}{2} (z-w)^2 : \partial^2 b c : \right)$$

$$+ \lambda(\lambda-1) \left(\frac{1}{(z-w)} \right) \left(: \partial c(w) \partial b(w) : \right) + \lambda(\lambda-1) \frac{(-2)}{(z-w)^3}$$

$$(ic) = \frac{2\lambda(\lambda-1)}{(z-w)^3} \left(: b(w) c(w) : \right) - \frac{2\lambda(\lambda-1)}{(z-w)^2} : \partial b(w) c(w) :$$

$$+ \frac{\lambda(\lambda-1)}{(z-w)} \left(: \partial c(w) \partial b(w) : - : \partial^2 b(w) c(w) : \right)$$

$$= \frac{2\lambda(\lambda-1)}{(z-w)^4}$$

Okh... let's just calculate all $\sim \frac{1}{(z-w)^4}$ terms

$$|ii\rangle \lambda(\lambda-1) : \partial b(w) c(w) : : b(z) \partial c(z) :$$

$$\sim \lambda(\lambda-1) \langle \partial b(w) \partial c(z) \rangle \langle c(w) b(z) \rangle$$

$$= \lambda(\lambda-1) \frac{-2}{(z-w)^3} \frac{1}{(z-w)} = \lambda(\lambda-1) \left(\frac{-2}{(z-w)^4} \right)$$

$$|iv\rangle (1-\lambda)^2 : \partial b(z) c(z) : : \partial b(w) c(w) :$$

$$\sim (1-\lambda)^2 \langle \partial b(z) c(w) \rangle \langle c(z) \partial b(w) \rangle$$

$$= (1-\lambda^2) \left(\frac{-2}{(z-w)^2} \right) \left(\frac{1}{(z-w)^2} \right) = -(1-\lambda)^2 \frac{1}{(z-w)^4}$$

$$= \underline{(-1 + 2\lambda - \lambda^2) / (z-w)^4}$$

$$-1 + 2\lambda - \lambda^2 - 2\lambda^2 + 2\lambda = -2\lambda^2 + 2\lambda - 1$$

$$(-1 - 6\lambda^2 + 6\lambda) / (z-w)^4 = \frac{1}{2} c^{b,c} / (z-w)^4$$

and so

$$c^{b,c} = 2(-1 + 6\lambda - 6\lambda^2)$$

as claimed

* For the Bosonic string $\lambda = 2$:

$$c^{b,c} = -26$$

Highest weight states:

- Suppose A and $\bar{A}$ are the Virasoro alg. generated by $T(z)$ and $\bar{T}(\bar{z})$ respectively
- want to construct highest weight states of the Virasoro Algebra.

A highest weight state $|h\rangle$ corresponding to a primary field of cent. dimension h has

$$L_n |h\rangle = 0 \quad n > 0$$

$$L_0 |h\rangle = h |h\rangle$$

$$(i) L_n \phi_{-n} |0\rangle = ([L_n, \phi_{-n}] + \phi_{-n} L_n) |0\rangle$$

$$\lim_{z \rightarrow 0} T(z) |0\rangle = \sum_{n=-\infty}^{\infty} L_n z^{n-2} |0\rangle$$

$$n+2 > 0 \Rightarrow n+2 \geq 0$$

$$L_n |0\rangle = 0 \text{ for } n \geq -2$$

$$\text{or for } n > 0$$

$$(ii) L_n \phi_{-n} |0\rangle = ([L_0, \phi_{-n}] + \phi_{-n} L_n) |0\rangle \quad (n > 0)$$

$$= [L_0, \phi_{-n}] |0\rangle$$

$$= (h(n+1) - n) \phi_{-n} |0\rangle$$

$$n=0$$

$$L_0 |h\rangle = h |h\rangle \quad \checkmark$$

Verma module: $V_{h,c} = \{ L_{k_1} \dots L_{k_n} |0\rangle ; k_i \leq -2 \}$
w/ central charge c

• lowest level states:

$$L_{-1} |h\rangle, L_{-2} |h\rangle, L_{-1} L_{-1} |h\rangle, L_{-3} |h\rangle$$

$V^{h,c} \sim$ Set of states corresponding to the conf. family $[\phi(z)]$ of a primary field $\phi(z)$ w conf. dim h

Application to String theory:

World sheet coordinates $(\tau, \sigma) : \Sigma_{(2,1)}$
 $\leadsto$ light-cone coords $\xi_{\pm} \equiv \tau \pm \sigma$

Closed string: map $\Sigma_{(2,1)} \cong S^1 \times \mathbb{R} \rightarrow \mathbb{CP}^1$

$$\bullet -\infty < \tau < \infty, -\pi < \sigma \leq \pi$$

with

$$X^M(\sigma + 2\pi) = X^M(\sigma)$$

(i) Wick rotate to $\mathbb{E}^2$

$$\tau \rightarrow -i\tau$$

$$\xi^\pm \rightarrow -i\tau \pm \sigma = -i|\tau \pm i\sigma|$$

$$\xi'^\pm = -i(\tau \pm i\sigma)$$

• Now define complex coordinates

$$z \equiv \tau + i\sigma \quad | \quad \bar{z} \equiv \tau - i\sigma$$

with the sides $\sigma = -\pi$ and $\sigma = \pi$ identified

i.e.

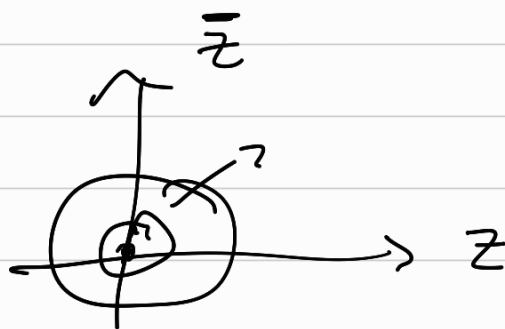
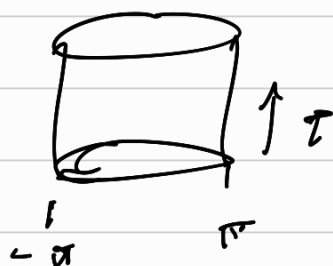
$$z' = \tau + i(\sigma + 2\pi) = \tau + i\sigma + 2\pi i$$

$$\sim \tau + i\sigma = z$$

$$z' \sim z \Leftrightarrow z' - z = 2\pi i$$

• These are $\mathbb{C}$ coordinates on $S^1 \times \mathbb{R}$

Now map



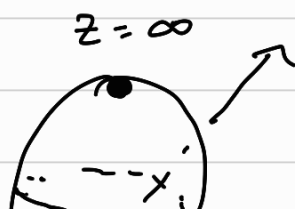
$$z = \tau + i\sigma = \ln(z) \quad | \quad z = e^z = e^{\tau + i\sigma}$$

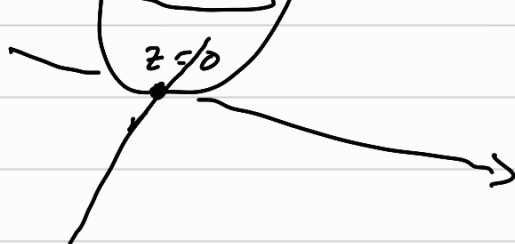
$$\bar{z} = \tau - i\sigma = \ln(\bar{z}) \quad | \quad \bar{z} = e^{\bar{z}} = e^{\tau - i\sigma}$$

• Open string: $-\infty < \tau < \infty$, $0 \leq \sigma \leq \pi$

* $z = 0$ is the south pole of $S^1 \cup \{\infty\}$

$z = \infty$ is the N-pole of $S^1 \cup \{\infty\}$





- Time - translation : $\tau' = \tau + t$ on $S^1 \times \mathbb{R}$
 $\sigma' = \sigma$

$$z \rightarrow z' = e^t z$$

* Becomes a Dilation of $\mathbb{CP}^1$ generated by $L_0 + \bar{L}_0$ (closed)

{ * Generators $-z^{n+1} L_n \Rightarrow z L_0 \sim z \frac{\partial}{\partial z}$
 * Gen. by L_0 for open string.

- Primary field R w weight $(\lambda, \bar{\lambda})$

$$R(z, \bar{z}) = z^\lambda \bar{z}^{\bar{\lambda}} R(\zeta, \bar{\zeta})$$

$$R(z, \bar{z}) = \left(\frac{d\omega}{dz}\right)^\lambda \left(\frac{d\bar{\omega}}{d\bar{z}}\right)^{\bar{\lambda}} R(\omega, \bar{\omega})$$

$$\omega(z) = \zeta = \lambda_1(z)$$

$$\bar{\omega}(\bar{z}) = \bar{\zeta} = \bar{\lambda}_1(\bar{z})$$

$$R(z, \bar{z}) = \left(\frac{1}{z}\right)^\lambda \left(\frac{1}{\bar{z}}\right)^{\bar{\lambda}} R(\zeta, \bar{\zeta})$$

$$\Rightarrow z^\lambda \bar{z}^{\bar{\lambda}} R(z, \bar{z}) = R(\zeta, \bar{\zeta})$$

as claimed

$$S^1 \times \mathbb{R} : R(\zeta) = \sum R_n e^{-n\zeta}$$

$$\text{and so } R(z) = z^{-\lambda} R(\zeta)$$

$\mathbb{CP}^1$

$$R(z) = z^{-\lambda} \sum R_n e^{-n \ln(z)} \\ = \sum R_n z^{-\lambda} z^{-n}$$

$$\text{or } R(z) = \sum R_n z^{-n-\lambda}$$

which is exactly why we use this expansion in $z^{-n-\lambda} \dots$

* String field : $x^\mu(\tau, \sigma) : (t, \bar{t}) = (0, 0)$

Closed String :

$$X^{\mu}(z, \bar{z}) = X_L^{\mu}(z) + X_R^{\mu}(\bar{z})$$

$$\begin{aligned} X_L^{\mu}(z) &= q_L^{\mu} - i \frac{\alpha'}{2} P_L^{\mu} \ln(z) + i \sqrt{\frac{\alpha'}{2}} \sum_{n \neq 0} \frac{1}{n} \alpha_n^{\mu} z^{-n} \\ X_R^{\mu}(\bar{z}) &= q_R^{\mu} - i \frac{\alpha'}{2} P_R^{\mu} \ln(\bar{z}) + i \sqrt{\frac{\alpha'}{2}} \sum_{n \neq 0} \frac{1}{n} \bar{\alpha}_n^{\mu} \bar{z}^{-n} \\ q^{\mu} &= q_L^{\mu} + q_R^{\mu}, \quad P^{\mu} = \frac{1}{2}(P_L^{\mu} + P_R^{\mu}) \end{aligned}$$

we generally take $\alpha_0^{\mu} = \sqrt{\alpha'/2} P^{\mu} = \bar{\alpha}_0^{\mu}$
 $P_L^{\mu} = P_R^{\mu}$

• Closed string: $\alpha' = 2$

~> This is simply D-copies of the scalar field φ , $\mu = 1, \dots, D$.

=> X^{μ} has all the same CFT properties as φ : just add the index μ .

• States and scattering involve the operator $:\exp[i k^{\mu} X_{\mu}(z, \bar{z})]:$

Open String: Given on the boundary $\sigma=0$

$$Q^{\mu}(z) = q^{\mu} - i \sqrt{2\alpha'} \alpha_0^{\mu} \ln(z) + i \sqrt{2\alpha'} \sum_{n \neq 0} \frac{1}{n} \alpha_n^{\mu} z^{-n}$$

$$z = e^{\tau} \mid \alpha_0^{\mu} = \sqrt{2\alpha'} P^{\mu} \quad \hookrightarrow \sim \Sigma(1) \cos(m\sigma)$$

~> For all σ :

$$X^{\mu}(z) = q^{\mu} - i \sqrt{\frac{\alpha'}{2}} \alpha_0^{\mu} (\ln(z) + \ln(\bar{z})) + i \sqrt{\frac{\alpha'}{2}} \sum_{n \neq 0} \left(\frac{1}{n} \alpha_n^{\mu} z^{-n} + \frac{1}{n} \bar{\alpha}_n^{\mu} \bar{z}^{-n} \right)$$

however

$$\dot{X}^{\mu} = \sqrt{\frac{2}{\alpha'}} i \partial X^{\mu} = \frac{i}{\sqrt{2\alpha'}} \partial Q^{\mu} = \sum_{n=0}^{\infty} \alpha_n^{\mu} z^{-n}$$

=> Recall $j^{\mu}(z) = i \partial X^{\mu}(z, \bar{z})$ is a conserved (chiral) current with weight $(1, 0)$

$$T = -\frac{1}{2\sqrt{2\alpha'}} : \partial Q^{\mu} \partial Q^{\nu} \eta_{\mu\nu} :$$

$$\sqrt{2/\alpha'} j^{\mu} = \frac{i}{\sqrt{2\alpha'}} \partial Q^{\mu} \Rightarrow j^{\mu} = \frac{i}{2} \partial Q^{\mu}$$

$$Q(z)Q(w) = :Q(z)Q(w): - 2\eta^{\mu\nu}\alpha' \chi_1(z-w)$$

and so

$$\partial Q^\mu(z) \partial Q^\nu(w) = -2\alpha' \eta^{\mu\nu} \frac{1}{(z-w)^2} + \dots$$

Replace $\varphi_L \rightarrow Q$

* T is not a primary field but

$$T_{\text{eye}}(\zeta) = \zeta^2 T_{\text{app}}(z) - (c/24)$$

→ In chp 3: Quantized Bosonic string using BRST ghost-fields.

$$\begin{array}{l|l} b_{\zeta\bar{\zeta}} : (2,0) & b_{\bar{\zeta}\zeta} : (0,2) \\ c^{\bar{\zeta}} : (-1,0) & c^{\zeta} : (0,-1) \end{array}$$

Bosonic Ghost Action:

$$S = -\frac{1}{4\pi\alpha'} \int d^2\zeta \eta^{\alpha\beta} \partial_\alpha X^\mu \partial_\beta X^\nu \eta_{\mu\nu} + \frac{1}{\pi} \int d^2\zeta b_{\alpha\beta} (\partial^\alpha c^\beta + \partial^\beta c^\alpha \eta^{\alpha\beta}) \quad \partial_\delta c^\delta$$

$$\zeta^\pm = \tau \pm \sigma, \quad \bar{\zeta} = -i(\tau + i\sigma)$$

$$d\zeta^+ d\bar{\zeta}^- = \frac{i}{2} d\zeta d\bar{\zeta}$$

$$\partial_+ \rightarrow i\partial_\zeta, \quad \partial_- \rightarrow i\partial_{\bar{\zeta}}$$

• Rescale

$$\begin{array}{l} X^\mu \rightarrow \frac{1}{2} X^\mu, \quad c^{\bar{\zeta}} \rightarrow c^{\bar{\zeta}}, \quad b_{\bar{\zeta}\zeta} \rightarrow -\frac{1}{4} b_{\bar{\zeta}\zeta} \\ c^{\zeta} \rightarrow c^{\zeta}, \quad b_{\zeta\bar{\zeta}} \rightarrow -\frac{1}{4} b_{\zeta\bar{\zeta}} \\ 2\alpha' = 1, \\ b_{\zeta\zeta} \rightarrow i b_{\zeta\zeta} \end{array}$$

we obtain:

$$S = -\frac{1}{2\pi} \int d\zeta d\bar{\zeta} \left\{ -\frac{1}{2} \partial_\zeta X^\mu \partial_{\bar{\zeta}} X^\nu \eta_{\mu\nu} - (b_{\zeta\bar{\zeta}} \partial_{\bar{\zeta}} c^\zeta + b_{\bar{\zeta}\zeta} \partial_\zeta c^{\bar{\zeta}}) \right\} \quad \partial_\zeta c^{\bar{\zeta}}$$

→ Found

$$T_{\alpha\beta} = \partial_\alpha X^\mu \partial_\beta X_\mu - \frac{1}{2} \eta_{\alpha\beta} \partial_\gamma X^\mu \partial^\gamma X_\mu + 4\alpha' \{ (b_{\alpha\delta} \partial_\beta c^\delta + b_{\beta\delta} \partial_\alpha c^\delta) + (\partial_\gamma b_{\alpha\beta}) c^\gamma \}$$

$$\text{replace } \eta_{\alpha\beta} \rightarrow g_{\alpha\beta} = \begin{pmatrix} 0 & 1/2 \\ 1/2 & 0 \end{pmatrix}$$

$$\tilde{T}_{\zeta\zeta} = \partial_\zeta X^\mu \partial_\zeta X_\mu + 4 \{ b_{\zeta\zeta} \partial_\zeta c^{\bar{\zeta}} \} + 2 \partial_\zeta b_{\zeta\zeta} c^{\bar{\zeta}}$$

Rescale by $-1/2$: $T_{\bar{z}\bar{z}} = -\frac{1}{2} \tilde{T}_{\bar{z}\bar{z}}$

$$T_{\bar{z}\bar{z}} = -\frac{1}{2} \partial_{\bar{z}} x^{\mu} \partial_{\bar{z}} x_{\mu} - 2 b_{\bar{z}\bar{z}} c^{\bar{z}} - (\partial_{\bar{z}} b_{\bar{z}\bar{z}}) c^{\bar{z}}$$

$T_{\bar{z}\bar{z}} \rightarrow \bar{z} \rightarrow \bar{z}$ in $T_{\bar{z}\bar{z}}$.

* Can now map all quantities to $\mathbb{CP}^1$

$$b_{zz} = z^{-2} b_{\bar{z}\bar{z}}(\bar{z})$$

$$b_{zz} = \sum_n b_n z^{-n-2}$$

$$A = \frac{i}{4\pi} \int dz d\bar{z} \left[\frac{1}{2} \partial x^{\mu} \bar{\partial} x^{\nu} \eta_{\mu\nu} + (b \bar{\partial} c + \bar{b} \partial \bar{c}) \right]$$

$$\begin{aligned} T &= -\frac{1}{2} \partial x^{\mu} \partial x^{\nu} \eta_{\mu\nu} - 2 b (\partial c) - (\partial b) c \\ \bar{T} &= -\frac{1}{2} \bar{\partial} x^{\mu} \bar{\partial} x^{\nu} \eta_{\mu\nu} - 2 \bar{b} (\bar{\partial} \bar{c}) - (\bar{\partial} \bar{b}) \bar{c} \end{aligned}$$

$$\begin{aligned} b &= b_{zz}, c = c^z \\ \bar{b} &= b_{\bar{z}\bar{z}}, \bar{c} = c^{\bar{z}} \end{aligned}$$

Open string: $0 \leq \sigma \leq \pi$

Recall $T^{b,c} = -\lambda : b \partial c : + (1-\lambda) : \partial b c :$

so $\lambda = 2$

T : E.M. tensor of D-scalars (0,0) and a first order ghost system with $\lambda = 2$.

$\Rightarrow$ Has central charge $c = D - 26$

Recall:

$$T^{\psi} T^{\psi} = \frac{1}{2(z-w)^4} + \dots$$

so if we now have D-scalars:

$$\Rightarrow R(TT) (\text{Scalar sector}) \sim \frac{D}{2(z-w)^4}$$

$$\Rightarrow R(TT) (\text{ghost system}) \sim \frac{c_{\text{ghost}}}{2(z-w)^4}$$

$$c^{b,c} = 2(-1 + \mathcal{O}(2)) = \mathcal{O}(2)^2$$

$$= 2(-1 + 12) = 24 = -26$$

$$c^h = -26 \quad (D-26)$$

$$\therefore T(z)T(w) = \frac{(D-26)}{2(z-w)^4} + (\dots) + \dots$$

* The central charge is an anomaly in the conformal symmetry

→ Since

$$T_{\text{reg}}(z) = T_{\text{op}}(z) z^2 - \frac{c}{24} \left(\frac{2\pi}{l} \right)^2$$

If we have the vacuum en. density

$$\langle T_{\text{op}}(z) \rangle = 0$$

then

$$\langle T_{\text{reg}}(z) \rangle = -\frac{c}{24} \left(\frac{2\pi}{l} \right)^2$$

→ which should vanish and hence leads to

$$\text{inequivalent vacuum}$$

$$\langle \phi(z_1) \dots \phi(z_n) \rangle = \sum_{i=1}^n \oint_{z_i} \frac{dz}{2\pi i} \xi \langle T(z) \phi_1 \dots \phi_n \rangle$$

$$(\phi_1 \dots \phi_n) = \mathbb{1}$$

$$\text{LHS} = 0$$

$$\therefore \langle T(z) \rangle = 0$$

Recall

$$\langle \phi_i(z) \phi_j(w) \rangle = \frac{\delta_{ij} \alpha}{(z-w)^{2h}} \quad \alpha : \text{const.}$$

$$T: h=2$$

$$\langle T(z) T(w) \rangle = \frac{c/2}{(z-w)^4}$$

$$\therefore \langle T(z) T(0) \rangle = c/2 / z^4$$

$$\begin{aligned} \langle \delta T(0) \rangle &= \int \frac{dz}{2\pi i} \xi(z) R[T(z) T(0)] \\ &= \int \frac{dz}{2\pi i} \xi(z) (c/2) / z^4 \\ &= c/2 \partial^3 \xi(z) \end{aligned}$$

However for a cont. trans $\xi \sim \mathcal{O}(z^2)$ at

most go RHS vanishes

→ Conditions for cont. trans.

$$\begin{aligned} \langle T(w) T(z) \rangle &= \left(\frac{2}{(w-z)^2} + \frac{1}{(w-z)} \partial \right) \langle T(z) T(w) \rangle \\ &\quad + c/2 / (z-w)^4 \end{aligned}$$

$$\langle T(w) T(0) \rangle = \left(\frac{2}{(w)^2} + \frac{1}{4} \frac{c}{w^2} \right) \langle T(w) T(0) \rangle + c/2 (w)^4$$

• For c to vanish we must have $\boxed{D=26}$

$$(\text{BRST charge}) Q = \oint_0 \frac{dz}{2\pi i} c(z) \left(T^x(z) + \frac{1}{2} T^{bc}(z) \right)$$

$$\{Q(z), Q(w)\} = \oint_{\Gamma_z} \frac{dz}{2\pi i} \oint_{\Gamma} \frac{dw}{2\pi i} \left\{ c(z) \left(T^x(z) + \frac{1}{2} T^{bc}(z) \right) \right\} \quad "w"$$

$$= 0 \quad \text{iff} \quad D=26$$

$\Rightarrow$ i.e. there is no BRST alg. for $D=26$ and hence there are no ghost states in the spectrum.

$$Q_B^2 = 0$$

Physical States: $|\psi\rangle$ obey

$$(L_0 - 1)|\psi\rangle = 0$$

$$L_n |\psi\rangle = 0, \quad n \geq 1$$

Thus, physical states are highest weight states with weight $1 = h$.

Tachyon state: $|0, p\rangle, \quad p^2 = 2$

$\Rightarrow$ Associated primary field: $: e^{i p_\mu Q^\mu(z)} :$

has weight $h_Q = 1$

$$e^{i p Q_2(z)} e^{i p_4 2\alpha' p^2 \ln(z)} e^{i p Q_3(z)} \quad \left| \begin{array}{l} Q_1 = -i \sum_n \frac{\alpha_n}{n} z^n \\ Q_3 = i \sum_n \frac{\alpha_n}{n} z^{-n} \end{array} \right.$$

$$\lim_{z \rightarrow 0} e^{i p Q} e^{i p_4 2\alpha' p^2 \ln(z)} e^{i p Q_3(z)} |0, 0\rangle$$

$$\text{since } \alpha_n^\mu |0\rangle = 0 \quad \forall n$$

$$|0, 0\rangle :$$

(Notes from Polchinski):

$|0, k\rangle$, $k = (k^0, k^i)$ is defined to be annihilated by lowering ops:

$$\hat{p}^0 |0, k\rangle = k^0 |0, k\rangle$$

$$\hat{p}^i |0, k\rangle = k^i |0, k\rangle$$

$$\alpha_m^i |0, k\rangle = 0, m > 0$$

* $|0, k\rangle$: Has mass $m^2 = \frac{2-D}{24\alpha'}$
 so $m^2 < 0$ for $D > 2$

$$\mathcal{H} = |0, 0\rangle \oplus \mathcal{H}_1 \oplus \mathcal{H}_2 \oplus \dots$$

$$\cong |0, k\rangle$$

• Single string with momentum k .

$$|0, k\rangle = \lim_{z \rightarrow z_0} \left(e^{i k^\mu x_\mu(z, \bar{z})} |0, 0\rangle \right)$$

To understand this: $x^\mu \rightarrow x^\mu + a^\mu$
 $:e^{i k^\mu x_\mu}: \rightarrow :e^{i k^\mu a_\mu}:$

The State

$$|0, k\rangle \rightarrow e^{i k a} |0, k\rangle$$

$$\langle k, 0 | 0, k' \rangle = 2\pi \delta(k - k')$$

$$\alpha_n^\mu |0, p\rangle = 0 \quad n \geq 1$$

$$\alpha_0^\mu |0, p\rangle = \sqrt{2\alpha'} p^\mu |0, p\rangle$$

Claim: $e^{i k^\mu q_\mu} |0\rangle = |k\rangle$

q_μ : Position operator

↪ Back to (0.3.15):

Vacuum: $|0, 0\rangle$

1PS (1 Particle State): $|0, p\rangle$

so α_0^μ

Thus: $\alpha_n^\mu = \alpha^\mu p_{n+1} + i \sum_{m \neq -n} \frac{1}{n-m} \alpha^\mu p_{n-m}$

$$\lim_{z \rightarrow 0} \left(: e^{i p^\mu Q_\mu(z)} : |0,0\rangle \right)$$

$$\exp(i p^\mu \left(-i \sum_{n=1}^{\infty} \frac{\alpha_{-n}}{n} z^n \right)) e^{i p^\mu (q_\mu - i k_\mu \ln(z))}$$

$$\times \exp(i p^\mu \left(i \sum_{n=1}^{\infty} \frac{\alpha_n}{n} z^{-n} \right)) |0,0\rangle$$

→ Gives:

$$1 \text{ since } \alpha_n^+ |0\rangle = 0, n > 0$$

$$\text{Gives } 1 \lim_{z \rightarrow 0} () = 1$$

Only left with

$$\lim_{z \rightarrow 0} e^{i p^\mu (q_\mu - i \alpha_0^+ \ln(z))} |0,0\rangle$$

$$\alpha_0^+ |0,0\rangle = 0$$

and so we are left with

$$e^{i p^\mu q_\mu} |0,0\rangle = |0,p\rangle$$

or

$$: e^{i p^\mu Q_\mu(z)} : |0,0\rangle = |0,p\rangle$$

$$p^2 = 2$$

Mass operator: $\alpha' \tilde{M}^2 = : \sum_{n=0}^{\infty} \alpha_{-n}^\mu \alpha_{\mu n} : - 1 = -p^2$

$$N=0: \alpha' M^2 = -1 \rightarrow |0,k\rangle$$

$$N: \alpha' \tilde{M}^2 = N - 1 = -p^2 \rightarrow \alpha_{-1}^\mu |0,k\rangle$$

$$N=0: |0,k\rangle: \alpha' M^2 = -1$$

$$\text{but } 2\alpha' = 1 \text{ so } M^2 = -2 = -p^2$$

and so this is why $p^2 = 2$. As such

$$: e^{i p^\mu Q_\mu(z)} : \text{ has } h = 1$$

$$\Rightarrow |4\rangle \text{ s.t. } L_0 |4\rangle = |4\rangle$$

$$|4\rangle = \lim_{z \rightarrow 0} \left(: e^{i p_M Q_M(z)} : |0,0\rangle \right) = |0,4\rangle$$

* $N=1$: will have: $\alpha_{-1}^\mu |0,p\rangle$ (Vector Boson)

and since $\partial Q^\mu = \sum_n \alpha_n^\mu z^{-n}$ we can

use this to look at

$$: \epsilon_\mu \partial Q^\mu e^{i k^\mu Q_\mu(z)} :$$

Claim: $h=1$ $\checkmark$

proof:

$$T(z) : \epsilon_\mu \partial Q^\mu(w) e^{i k Q(w)} : = (\text{veg}) + \dots$$

If we use:

$$\text{Where } T = -\frac{1}{2} \partial Q^\mu \partial Q^\nu \eta_{\mu\nu}$$

$$R(\partial Q^\mu(z) \partial Q^\nu(w)) = -\eta^{\mu\nu} \frac{1}{(z-w)^2} + : \partial Q^\mu(z) \partial Q^\nu(w) :$$

$$(-\frac{1}{2} \partial Q^\mu(z) \partial Q^\nu(z) \eta_{\mu\nu}) : \epsilon_\sigma \partial Q^\sigma(w) e^{i k Q(w)} :$$

$$= (\text{veg}) - \langle \partial Q^\mu(z) \partial Q^\sigma(w) \rangle : \partial Q_\mu(z) e^{i k Q(w)} : \epsilon_\sigma$$

$$- \langle \partial Q^\mu(z) e^{i k Q(w)} \rangle : \partial Q_\mu(z) \epsilon_\sigma \partial Q^\sigma :$$

$$- \langle \partial Q^\mu(z) \partial Q^\sigma(w) \rangle \langle \partial Q^\nu(z) e^{i k Q} \rangle \epsilon_\sigma \eta_{\mu\nu}$$

$$(i) \eta^{\mu\sigma} \frac{1}{(z-w)^2} : \epsilon_\sigma \partial Q_\mu(z) e^{i k Q(w)} :$$

$$= \frac{1}{(z-w)^2} \epsilon_\mu : \partial Q^\mu(w) e^{i k Q(w)} : + \frac{1}{z-w} \epsilon_\mu : \partial^2 Q_\mu(w) e^{i k Q(w)} :$$

$$(ii) \langle \partial Q^\mu(z) \sum_{n=1}^{\infty} \frac{(i k Q(w))^n}{n!} \rangle$$

$$\langle \partial Q^\mu(z) : k Q(w) \rangle : \partial Q_\mu(z) \epsilon_\sigma \partial Q^\sigma(w) e^{i k Q(w)} :$$

$$\left(\frac{i k^\mu}{(z-w)} \right) : \partial Q_\mu(w) \epsilon_\sigma \partial Q^\sigma(w) e^{i k Q(w)}$$

$$(iii) \eta^{\mu\sigma} / (z-w)^2 \left(\frac{-i k}{(z-w)} \right) e^{i k Q(w)} \epsilon_\sigma \eta_{\mu\nu}$$

$$\sim \left(\eta^{\mu\sigma} / (z-w)^2 \right) \left(\frac{-i k_\lambda \eta^{\nu\lambda}}{(z-w)} \right) e^{i k Q} \epsilon_\sigma \eta_{\mu\nu}$$

$$= i k^\sigma \epsilon_\sigma e^{i k Q(z)} / (z-w)^3$$

→ Missing the following contraction

$$\begin{aligned} -\frac{1}{2} \partial Q^M(z) \partial Q^N(z) e^{i k Q(z)} &= -\frac{1}{2} \partial Q^M(z) \partial Q^N(z) : \sum_{n=0}^{\infty} \frac{(i k Q(z))^n}{n!} \\ &= n \langle \partial Q^M(z) i k^\sigma Q_\sigma(w) \rangle \langle \partial Q^N(z) \sum_{n=0}^{\infty} \frac{(i k Q(w))^{n-1}}{(n-1)!} \rangle \\ &= \langle \partial Q^M(z) i k^\sigma Q_\sigma(w) \rangle \langle \partial Q^N(z) i k^\tau Q_\tau(w) \rangle : \sum_{n=2}^{\infty} \frac{(i k Q(w))^n}{(n-2)!} \end{aligned}$$

in sum: $M = N = 2$

$$\begin{aligned} &= \frac{1}{2} \left(\frac{-i k}{(z-w)} \right)^2 : e^{i k Q(w)} \partial Q^\sigma(w) \epsilon_\sigma : \\ &= \frac{1}{2} \left(\frac{k^2}{(z-w)^2} \right) : \epsilon_\sigma \partial Q^\sigma(w) e^{i k Q(w)} : \frac{1}{(z-w)^2} \end{aligned}$$

$$Q(z-w)^{-3} : i : \epsilon_\mu k^\mu e^{i k Q} : / (z-w)^3$$

$$Q(z-w)^{-2} : \frac{k^2}{2} : \epsilon_\sigma \partial Q^\sigma(w) e^{i k Q(w)} :$$

$$\begin{aligned} Q(z-w)^{-1} : \frac{1}{z-w} \epsilon_\mu : \partial^2 Q^\mu(w) e^{i k Q(w)} + \frac{i k^\mu}{(z-w)} : \partial Q_\mu(w) \epsilon_\sigma \partial Q^\sigma(w) : \\ = \frac{1}{(z-w)} \partial : \epsilon_\mu Q^\mu(w) e^{i k Q(w)} : \end{aligned}$$

⇒ Has conformal weight 1 if $\epsilon_\mu k^\mu = 0$
 $k^2/2 = 0$

$$\begin{aligned} \text{As such: } \lim_{z \rightarrow 0} (i \epsilon_\mu \partial Q^\mu(z) e^{i k Q(z)} : |0,0\rangle) \\ = \lim_{z \rightarrow 0} (\epsilon_\mu \partial Q^\mu(z)) \lim_{z \rightarrow 0} (e^{i k Q(z)} |0,0\rangle) \\ = \lim_{z \rightarrow 0} (i \epsilon_\mu \partial Q^\mu(z) |0,k\rangle) \\ = \lim_{z \rightarrow 0} (\epsilon_\mu \partial Q^\mu(z) |0,k\rangle) \end{aligned}$$

I think best is wrong.

$$\begin{aligned} \partial Q^\mu(z) &= -i \alpha_0^\mu \frac{1}{z} - i \sum_{n \neq 0} \alpha_n^\mu z^{-n-1} \\ &= -i \left(\alpha_0^\mu z^{-1} + \sum_{n \neq 0} \alpha_n^\mu z^{-n-1} \right) \\ &= -i \sum_n \alpha_n^\mu z^{-n-1} \end{aligned}$$

yup.

$$\Rightarrow i \partial Q^\mu(z) = \sum_n \alpha_n^\mu z^{-n-1}$$

So I think it should be

$$\lim_{\epsilon \rightarrow 0} |\epsilon_\mu : \partial Q^\mu(z) | 0, u \rangle = \epsilon_\mu \alpha^\mu_{-1} | 0, u \rangle$$

- This is exactly the state of a Spin 1 - Gauge field w/ polarization vector ϵ_μ .

Construction of string theories:

- (i) Take 2D-conformally invariant theory and coupled it to two-dimensional gravity.
- (ii) Demanded NO conformal anomaly: eq. to insisting the central charge was zero in the L and R moving sectors separately in the BRST quantised theory
- Central charge of $\{b, c\}$ and $\{\beta, \gamma\}$ must cancel.

Superstring: include ghosts $\{\beta, \gamma\}$ for the local

(RNS)

Susy

include ghosts $\{b, c\}$ for diffeomorphisms

$$\beta : (h, \bar{h}) = (3/2, 0)$$

$$\gamma : (h, \bar{h}) = (-1/2, 0)$$

- Λ : BRST parameter

↳ Replaces the local susy parameter when finding the BRST transformations

⇒ Λ must have the same index structure as the susy parameters.

- ghost action is independent of the details of the matter fields used and only depends on the sym. algebra of the matter fields that are BRST quant.

Superstring Symm	Ghost fields
(i) diffeomorphisms	$\{b, c\}$
(ii) Susy	$\{\beta, \gamma\}$

$$\{b, c\} : \zeta = -26$$

$$\{B, \gamma\} : C_B = 12$$

and so

$$C_{ghost} = C_b + C_B = 15$$

$\Rightarrow$ The matter content that carries a rep of conformal symmetry and have a central charge 15.

$\sim$ Closed String: L & R movers

- G_{SUSY}: Majorana-Weyl parameter ϵ_{SUSY}
- total central charge in both L & R moving sectors cancel to zero separately.

* Const. of the Superstring:

- free Bosons & fermions

Bosonic fields: X^μ , $\mu = 0, \dots, D-1$

ferm. fields: χ^μ

{ Recall: free Boson ϕ has $C_\phi = 1$
free fermion ψ has $C_\psi = 1/2$

Thus, if we have a total of D of each field the central charge of the matter field

$$C_{matter} = D + \frac{1}{2}D = \frac{3}{2}D$$

and so $C_{matter} = C_{ghost}$



$$\Rightarrow \frac{3}{2}D = 15 \Rightarrow$$

$$D = 10$$



* The Matter fields:

$$S_m = \frac{1}{4\pi} \int d\tau d\bar{\tau} \left[\partial X^\mu \bar{\partial} X^\mu - \chi^\mu \bar{\partial} \chi^\mu - \bar{\chi}^\mu \partial \bar{\chi}^\mu \right] \eta_{\mu\nu}$$

• obtain T from translational invariance

$$\delta X^\mu = a^\mu \partial_\lambda X^\mu = a^\tau \partial X^\mu + a^{\bar{\tau}} \bar{\partial} X^\mu$$

$$\sim \text{Focus only on } (a = a^\tau) \delta_\tau X^\mu = a \partial X^\mu$$

$\sim$ Iso late $(\bar{\partial})$

$$\textcircled{B} (\bar{\partial} a) (\partial X^\mu \partial X^\mu)$$

$$\textcircled{E} - \chi^\mu (\bar{\partial} c) \partial \chi^\mu$$

$$\sim \int d\tau d\bar{\tau} (\bar{\partial} c) \left[\partial X^\mu \partial X^\mu - \chi^\mu \partial \chi^\mu \right]$$

$$T_{zz} = -\frac{1}{2} : \dot{X} \dot{X} : = -\frac{1}{2} : \partial X^\mu \partial X_\mu : + \frac{1}{2} : \dot{X}^\mu \dot{X}_\mu :$$

* The supercurrent is defined by the susy param.

$$\delta_\epsilon X^\mu = i \bar{\epsilon} \dot{X}^\mu$$

$$\delta X^\mu = \partial X^\mu \epsilon = \sigma^\mu \partial_\lambda X^\mu \epsilon$$

=> look at ϵ term with $\bar{\partial} \epsilon$ since

$$\epsilon \bar{\partial} J(z) = 0$$

and so this will only come from the middle term

* The supercurrent satisfies

$$J_A^\alpha = -\frac{1}{2} (\rho^\beta \rho^\alpha \psi_\mu)_A \partial_\beta X^\mu$$

$$(\rho^\alpha)_A \rho^\beta J_B^\alpha = 0$$

which follows from $\rho_\alpha \rho^\beta \rho^\alpha = 0$

$$\rho^0 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

$$\rho^1 = \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}$$

$$\rho^\alpha \partial_\alpha = \begin{pmatrix} 0 & 2\partial \\ -\bar{\partial} & 0 \end{pmatrix}$$

$$- \dot{X}^\mu (\bar{\partial} \epsilon) 2 \partial X_\mu$$

$$\Rightarrow J(z) = 2(-2 \dot{X}^\mu \partial X_\mu)$$

$$J = -4 \partial X^\mu \dot{X}_\mu$$

* T and J are Quasi-primary fields:

* BRST: introduce Grassmann odd

parameters $\{b, c\}$, $\lambda = 2$

Grassmann even: $\{\beta, \gamma\}$, $\lambda = 3/2$

λ is given by the conformal weight of the corr. symmetry

$$T^{b,c} = : -2 b \partial c - \partial b c :$$

$$T^{\beta,\gamma} = : -\frac{3}{2} \beta \partial \gamma - \frac{1}{2} \partial \beta \gamma :$$

$$T^{ghst} = T^{b,c} + T^{\beta,\gamma}$$

Heterotic String: Built from the free Boson & fermion but treats the left and right sectors differently.

L : Superstring: $X_L^\mu \oplus \psi^\mu$, $\mu=0, \dots, 9$

R : Has no supersymmetry and is similar to the Bosonic string.

$\rightarrow$ has only $(\lambda=2, \{b, c\})$ $C_{b,c} = -26$

So we have 26 Bosonic fields

$$(X_R^\mu, \mu=0, \dots, 9 \oplus X_R^I, I=1, \dots, 16)$$

• Coordinates: $X^\mu = X_L^\mu + X_R^\mu$ make up the 10d spacetime of the theory.

• The remaining ψ^μ are associated with a Lie Alg. which must have rank 16 and be self dual.

$$\leadsto E_8 \oplus E_8 \text{ and } SO(32)/\mathbb{Z}_2$$

• Could have also considered $N=2$ susy instead of $N=1$

Here $\{Q_\alpha^I, Q_{\dot{\beta}}^{I\dot{5}}\} = - (C \gamma^\mu)_{\alpha\dot{\beta}} P_\mu \delta^{I\dot{5}} + (\text{Cent.})$
 $I, \dot{5} = 1, 2$

which has a $U(1)_R$ symmetry.

* When we couple the theory to gravity (to the gravity multiplet (e_a^μ, ψ_μ))

the theory has

(one e_a^μ , two ψ_α^μ , one A^μ)

corresponding to the $U(1)$ current.

$\leadsto$ Ghosts: $(\lambda=2, \{b, c\})$ (1) $\hookleftarrow$ Diffes
 $(\lambda=3/2, \{\beta, \delta\})$ (2) $\hookleftarrow$ Susy
 $(\lambda=1, \{b', c'\})$ (1) $\hookleftarrow$ $U(1)$

• We need (2) ~~two~~ $\{\beta, \delta\}$ ghost systems

since $N=2$.

$$C_{gh} = C_{b,c} + C_{\beta,\delta} + C_{b',c'} \\ = -26 + 2(11) - 2$$

$$C_{gh} = 6 \quad (N=2 \text{ Susy})$$

$$C_{b,c} = 2(-1+6-6) = -2$$

$$\lambda=1:$$

$$C_{b',c'} = 2(-1+6-6) = -2$$

* The simplest superconf. -mv. theory is given by two scalars A, B and two fermions $\psi_i, i=1,2$

$$C_{\text{matter}} = 2 + \frac{1}{2}(2) = 3$$

$$(A, B, \psi_i) \oplus (A', B', \psi'_i)$$

must have two - only in $D=2$.

~D Only need to invoke Lorentz symm for $D \geq 4$

ex: Could use $X^\mu, \mu=0, \dots, 3$

and then "fill in" the remaining central

charge $C_m = 22$ to counter the

string $C = -26$.

* Any (super) conformally invariant theory which has no (super)conf. anomaly, when coupled to sugra, provides a good string theory if it is modular invariant.